

3-DOF Mass-on-Rail Longitudinal Chain with Impact Excitation

Sanitized Academic Portfolio Report

Public release version

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Note: the program-code explanation section was removed from this public report copy.

INTRODUCTION:

Impulse-like excitations (short shocks) are common in rail-guided machinery, robotic slides, and deployable mechanisms. Although the input duration is brief, it can inject substantial energy and produce rich transient responses. This project aims to make these transient dynamics understandable and interpretable using a compact discrete model.

We model a **3-degree-of-freedom (3-DOF) Mass-on-Rail** system with generalized coordinates $\mathbf{x}(t) = [x_1 \ x_2 \ x_3]^T$. The equations of motion are written as

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}\dot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{f}_{ext}(t),$$

where $\mathbf{M}$ is the mass matrix, $\mathbf{K}$ the stiffness matrix, and $\mathbf{C}$ (if included) the damping matrix. When an impulse is applied to a given mass, multiple modes are activated with different participation levels; therefore, each mass response reflects the coupled system dynamics rather than a purely local effect.

The main project outputs include constructing $\mathbf{M}$ and $\mathbf{K}$, computing modal properties (natural frequencies and mode shapes), generating the impulse responses (x_1, x_2, x_3) , and comparing the effect of excitation location (Mass 1 vs Mass 3). In addition, the behavior is tracked through total mechanical energy to support interpretation of energy transfer and (when present) dissipation.

To deliver these results in a reproducible and presentation-ready way, we developed a **GUI-based analysis workbench**. The tool assembles the model from user inputs, runs modal analysis, produces time-domain responses, and reports (x_1, x_2, x_3) together with $E(t) = T(t) + V(t)$ in a single workflow, enabling clearer discussion beyond static plots.

Our Extensions (Beyond-Core Contributions)

In addition to the baseline modeling/analysis steps, we implemented extensions that improve realism and robustness:

- **Viscous damping (dashpot) support** via a global $\mathbf{C}$ matrix to capture realistic decay.
- **Coulomb (dry) friction** to represent practical dissipation in rail-guided systems.
- **Time-domain solver** for stable simulation when damping/friction is active.
- **Impulse interpretation option:** force pulse $f_{ext}(t)$ or equivalent initial velocity jump $\Delta v = \frac{F\Delta t}{m}$.
- **Real-time animation with basic controls** (speed scaling, zoom, restart) while updating (x_1, x_2, x_3) and $E(t)$.
- **State consistency and validation check** to prevent stale demos and non-physical or numerically risky runs.

The remainder of the report presents the model formulation, modal results, impulse-response simulations, and finally the implemented extensions and GUI features.

After the Introduction, the system's **matrix formulation (M–K–C)** is established and the **complete equations of motion** are derived; then, sensitivity studies on **mass**, **spring stiffness**, and **spring length** are conducted along with comparative analyses of **impulse intensity** and **impulse duration**, supported by tables and figures. Next, the effect of **excitation location** (which mass is impacted) is evaluated, and finally the response changes due to **viscous damping** and **Coulomb friction** are reported using comparative plots and summary tables.

- **Mode shapes:** Only computed for the **undamped** reference case (using $\mathbf{K}\phi = \omega^2\mathbf{M}\phi$). **No damp mode shapes** are computed when damping/friction is active.

We computed mode shapes from the **undamped linear** model using **M** and **K**. With damping, the same modes remain valid only under **proportional damping**; **general damping** requires **complex (state-space) modal analysis**. Since **Coulomb friction is nonlinear**, classical modal methods become only approximate, so damped/friction cases were solved in the time domain (RK4).

- **DE solver (time-domain):** For cases with **viscous damping** and/or **Coulomb friction**, the equations of motion are integrated in time using the **4th-order Runge–Kutta (RK4)** method.

-**Layout:** Wall - k0 - m1 - k1 - m2 - k2 - m3 - k3 - Wall

FORMULAS:

Lagrangian Equation = Kinetic Energy – Potential Energy

$$L = T \text{ (Kinetic Energy)} - V \text{ (Potential Energy)}$$

$$T = \frac{1}{2}m_1\dot{x}_1^2 + \frac{1}{2}m_2\dot{x}_2^2 + \frac{1}{2}m_3\dot{x}_3^2$$

$$V = \text{Potential Energy}$$

$$\text{For } k_0's \text{ Potential Energy} = \text{The amount of elongation belongs only to } x_1 = \frac{1}{2}k_0x_1^2$$

$$\text{For } k_1's \text{ Potential Energy} = \text{The amount of elongation is } (x_2 - x_1) = \frac{1}{2}k_1(x_2 - x_1)^2$$

$$\text{For } k_2's \text{ Potential Energy} = \text{The amount of elongation is } (x_3 - x_2) = \frac{1}{2}k_2(x_3 - x_2)^2$$

$$\text{For } k_3's \text{ Potential Energy} = \text{The amount of elongation belongs only to } x_3 = \frac{1}{2}k_3x_3^2$$

$$\text{Euler - Lagrange Equation} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_i} \right) - \frac{\partial L}{\partial x_i} = F_i$$

For the m_1 :

- Kinetic Part = $m_1 \ddot{x}_1$
- Potential Part = x_1 in both $k_0 (k_0 x_1)$ and $k_1 (-k_1 (x_2 - x_1))$
- Equation; $m_1 \ddot{x}_1 + k_0 x_1 - k_1 (x_2 - x_1) = 0$
- Final Equation = $m_1 \ddot{x}_1 + (k_0 + k_1) x_1 - k_1 x_2 = 0$

For the m_2 :

- Kinetic Part = $m_2 \ddot{x}_2$
- Potential Part = x_2 in both $k_1 (k_1 (x_2 - x_1))$ and $k_2 (-k_2 (x_3 - x_2))$
- Equation; $m_2 \ddot{x}_2 + k_1 (x_2 - x_1) - k_2 (x_3 - x_2) = 0$
- Final equation: $m_2 \ddot{x}_2 - k_1 x_1 + (k_1 + k_2) x_2 - k_2 x_3 = 0$

For the m_3 :

- Kinetic Part: $m_3 \ddot{x}_3$
- Potential Part = x_3 in both $k_2 (k_2 (x_3 - x_2))$ and $k_3 (k_3 x_3)$
- Equation; $m_3 \ddot{x}_3 + k_2 (x_3 - x_2) + k_3 x_3 = 0$
- Final Equation = $m_3 \ddot{x}_3 - k_2 x_2 + (k_2 + k_3) x_3 = 0$

MATRIX FORMS

Mass Matrix (M);

$$M = \begin{bmatrix} m_1 & 0 & 0 \\ 0 & m_2 & 0 \\ 0 & 0 & m_3 \end{bmatrix}$$

Stiffness Matrix (K);

$$K = \begin{bmatrix} k_0 + k_1 & -k_1 & 0 \\ -k_1 & k_1 + k_2 & -k_2 \\ 0 & -k_2 & k_2 + k_3 \end{bmatrix}$$

$$\det(K - \omega^2 M) = 0$$

$$\det \begin{bmatrix} (k_0 + k_1) - m_1 \omega^2 & -k_1 & 0 \\ -k_1 & (k_1 + k_2) - m_2 \omega^2 & -k_2 \\ 0 & -k_2 & (k_2 + k_3) - m_3 \omega^2 \end{bmatrix}$$

$$\rightarrow [(k_0 + k_1) - m_1 \omega^2] * [(k_1 + k_2) - m_2 \omega^2] * [(k_2 + k_3) - m_3 \omega^2] - (-k_2)(-k_2) - (-k_1) * [(-k_1)[(k_2 + k_3) - m_3 \omega^2] - (-k_2) * 0] = 0$$

$$\rightarrow [(k_0 + k_1) - m_1 \omega^2] * [(k_1 + k_2) - m_2 \omega^2] * [(k_2 + k_3) - m_3 \omega^2] - (k_2)^2 - (k_1)^2 [(k_2 + k_3) - m_3 \omega^2] = 0$$

Adding Damper to the Our System:

First Damper c_0 ; (connected to *the* m_1) (Just m_1 moving.) Velocity = $\dot{x}_1^2$

$$D_0 = \frac{1}{2} c_0 \dot{x}_1^2$$

Second Damper c_1 ; (between to the m_1 and m_2)

$$D_1 = \frac{1}{2} c_1 (\dot{x}_2 - \dot{x}_1)^2$$

Third Damper c_2 ; (between m_2 and m_3)

$$D_2 = \frac{1}{2} c_2 (\dot{x}_3 - \dot{x}_2)^2$$

Fourth Damper c_3 ; (connected to the m_3)

$$D_3 = \frac{1}{2} c_3 \dot{x}_3^2$$

Total Damper Energy Equation:

$$D_{total} = \frac{1}{2} c_0 \dot{x}_1^2 + \frac{1}{2} c_1 (\dot{x}_2 - \dot{x}_1)^2 + \frac{1}{2} c_2 (\dot{x}_3 - \dot{x}_2)^2 + \frac{1}{2} c_3 \dot{x}_3^2$$

Euler Lagrange Equation Form with Adding To the Damper:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_i} \right) - \frac{\partial L}{\partial x_i} + \frac{\partial D}{\partial \dot{x}} = F_i$$

For the m_1 damper effect:

- ✓ Take the derivative of the function D_{total} with respect to the terms containing $\dot{x}_1$
- Term 1 (c_0): $\frac{1}{2} c_0 \dot{x}_1^2 \rightarrow$ Derivative $\rightarrow c_0 \dot{x}_1$
- Term 2 (c_1): $\frac{1}{2} c_1 (\dot{x}_2 - \dot{x}_1)^2 \rightarrow$ Derivative $\rightarrow -c_1 (\dot{x}_2 - \dot{x}_1) = c_1 \dot{x}_1 - c_1 \dot{x}_2$
- Total Energy: $c_0 \dot{x}_1 + c_1 \dot{x}_1 - c_1 \dot{x}_2 = (c_0 + c_1) \dot{x}_1 - c_1 \dot{x}_2$

For the m_2 damper effect:

- ✓ Take the derivative of the function D_{total} with respect to the terms containing $\dot{x}_2$
 - Term 2 (c_1): $\frac{1}{2}c_1(\dot{x}_2 - \dot{x}_1)^2 \rightarrow \text{Derivative} \rightarrow c_1(\dot{x}_2 - \dot{x}_1)$
 - Term 3 (c_2): $\frac{1}{2}c_2(\dot{x}_3 - \dot{x}_2)^2 \rightarrow \text{Derivative} \rightarrow -c_2(\dot{x}_3 - \dot{x}_2)$
 - Total Energy: $c_1(\dot{x}_2 - \dot{x}_1) - c_2(\dot{x}_3 - \dot{x}_2) = -c_1\dot{x}_1 + (c_1 + c_2)\dot{x}_2 - c_2\dot{x}_3$

For the m_3 damper effect:

- ✓ Take the derivative of the function D_{total} with respect to the terms containing $\dot{x}_3$
 - Term 3 (c_2): $\frac{1}{2}c_2(\dot{x}_3 - \dot{x}_2)^2 \rightarrow \text{Derivative} \rightarrow c_2(\dot{x}_3 - \dot{x}_2)$
 - Term 4 (c_3): $\frac{1}{2}c_3\dot{x}_3^2 \rightarrow \text{Derivative} \rightarrow c_3\dot{x}_3$
 - Total Energy: $c_2(\dot{x}_3 - \dot{x}_2) + c_3\dot{x}_3 = -c_2\dot{x}_2 + (c_2 + c_3)\dot{x}_3$

Matrix Form of Damper:

$$C = \begin{bmatrix} c_0 + c_1 & -c_1 & 0 \\ -c_1 & c_1 + c_2 & -c_2 \\ 0 & -c_2 & c_2 + c_3 \end{bmatrix}$$

TOTAL EQUATION OF MOTION IN OUR SYSTEM:

$$\begin{bmatrix} m_1 & 0 & 0 \\ 0 & m_2 & 0 \\ 0 & 0 & m_3 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \end{Bmatrix} + \begin{bmatrix} c_0 + c_1 & -c_1 & 0 \\ -c_1 & c_1 + c_2 & -c_2 \\ 0 & -c_2 & c_2 + c_3 \end{bmatrix} \begin{Bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{Bmatrix} + \begin{bmatrix} k_0 + k_1 & -k_1 & 0 \\ -k_1 & k_1 + k_2 & -k_2 \\ 0 & -k_2 & k_2 + k_3 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$$

Correction note: the stiffness term multiplies the displacement vector $[x_1, x_2, x_3]^T$, not a velocity component.

Mode-Shape Computation (Undamped Reference):

Free vibration equation

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{0}$$

Harmonic assumption

$$\mathbf{x}(t) = \boldsymbol{\phi} e^{j\omega t}$$

Generalized eigenvalue problem

$$(\mathbf{K} - \omega^2 \mathbf{M})\boldsymbol{\phi} = \mathbf{0}$$

Natural frequencies

$$\det(\mathbf{K} - \omega^2 \mathbf{M}) = 0 \Rightarrow \omega_1, \omega_2, \omega_3$$

Mode shape for each mode i

$$(\mathbf{K} - \omega_i^2 \mathbf{M})\boldsymbol{\phi}^{(i)} = \mathbf{0}, \boldsymbol{\phi}^{(i)} = \begin{bmatrix} \phi_1^{(i)} \\ \phi_2^{(i)} \\ \phi_3^{(i)} \end{bmatrix}$$

Mass normalization

$$\boldsymbol{\phi}^{(i)} \leftarrow \frac{\boldsymbol{\phi}^{(i)}}{\sqrt{(\boldsymbol{\phi}^{(i)})^T \mathbf{M} \boldsymbol{\phi}^{(i)}}}$$

3-DOF IMPULSE RESPONSE: ROOT MASS SENSITIVITY STUDY

Subject: Full Spectrum Analysis of Impact Point Inertia ($m_1 = 0.5, 1.0, 2.0$ kg)

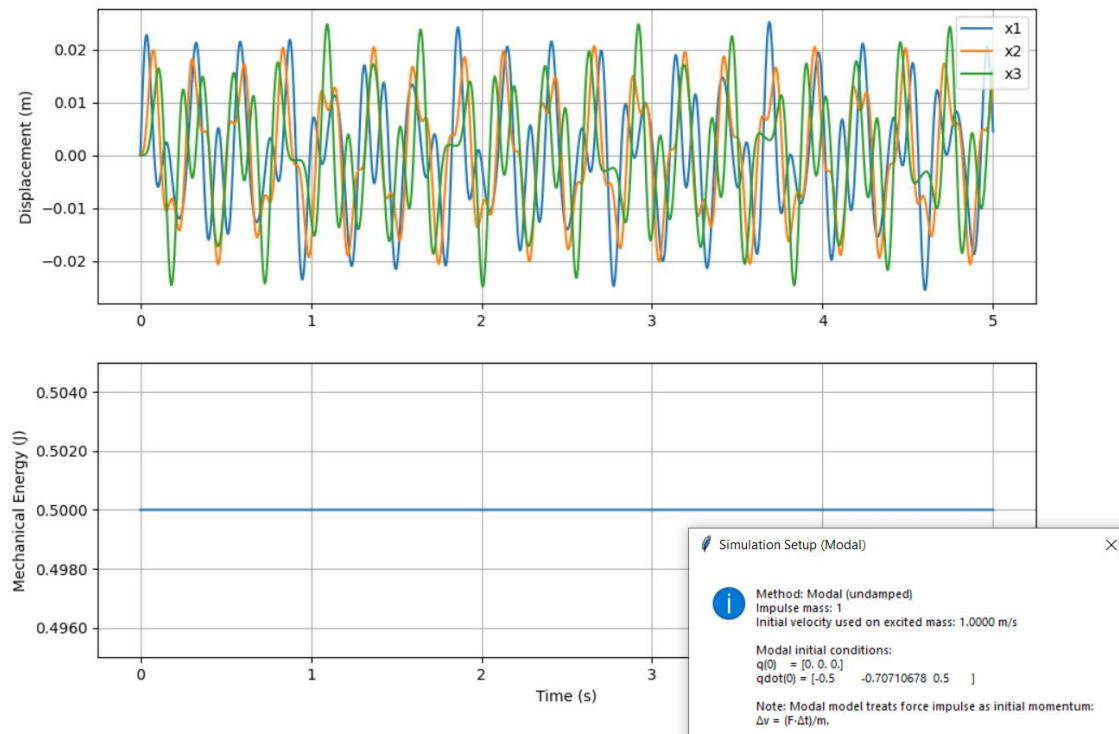


Figure 1: For $m_1 = 1.0$ kg

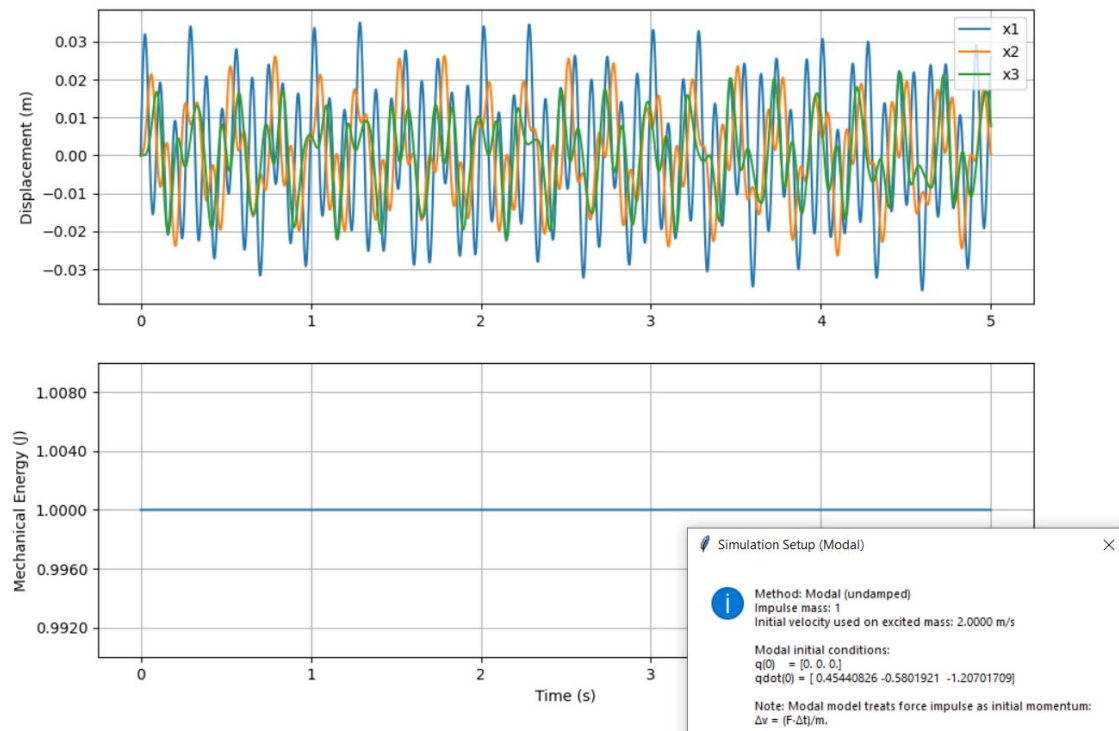


Figure 2: For $m_1 = 0.5$ kg

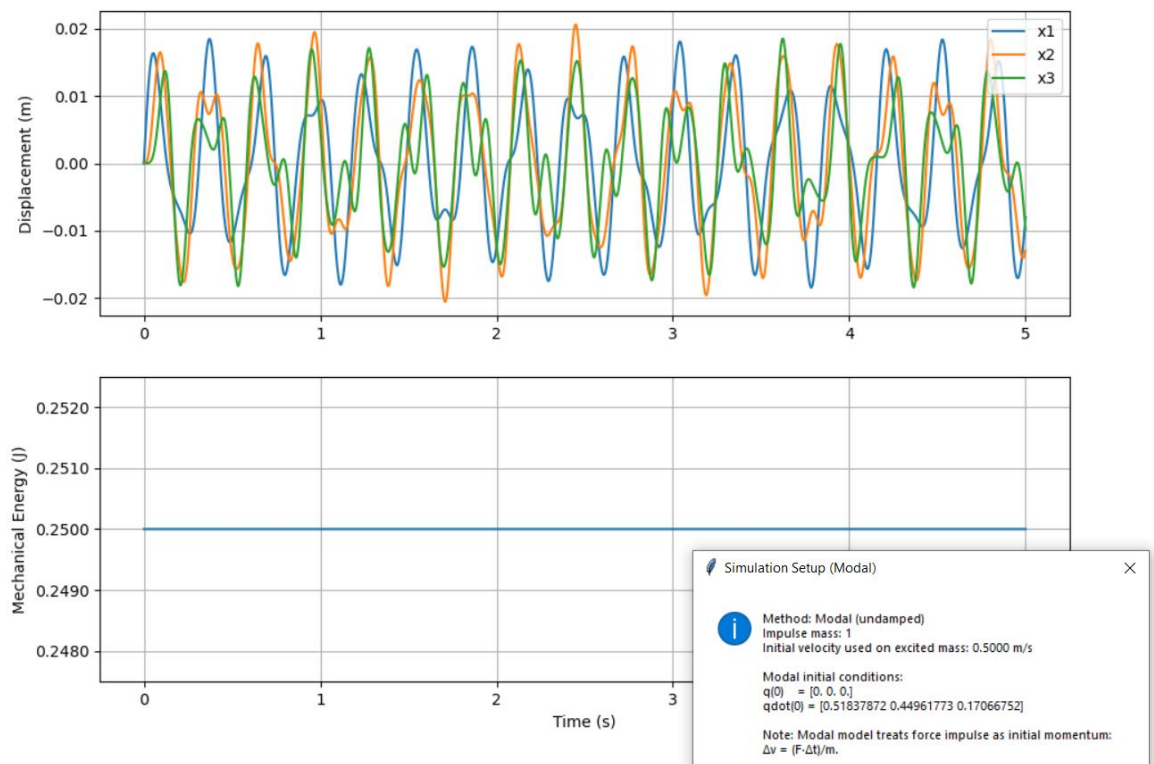


Figure 3: For $m_1 = 2.0 \text{ kg}$

COMPARATIVE TABLE 1

Feature	State S1 (0.5 kg)	State S2 (1.0 kg)	State S3 (2.0 kg)	Engineering Impact
Inertia Factor	0.5x (Low)	1.0x (Base)	2.0x (High)	Wide design range.
Input Energy	1.00 J (Max)	0.50 J	0.25 J (Min)	Inverse Law ($E \propto 1/m$)
Initial Velocity	2.0 m/s	1.0 m/s	0.5 m/s	Impact severity scales directly.
Disp. Profile	Root-Dominant ($x_1 \gg$)	Uniform ($x_1 \approx x_3$)	Root-Suppressed ($x_1 < x_2$)	Shift from "Driver" to "Anchor".
Response	Violent / High Freq.	Nominal	Controlled / Low Freq.	Frequency content filtering.

Comparative Analysis (The "Mass Effect" Spectrum):

This 3-point analysis reveals the fundamental physics of impulse admission:

1. Energy Inverse Law:

The system energy follows a perfect inverse relationship. Doubling the mass strictly halves the admitted energy ($1.0 \rightarrow 0.5 \rightarrow 0.25$ J). This confirms that for impulsive loads, **increasing mass is the most effective primary defense** against energy injection.

2. Mode Shape Transformation:

- **At 0.5 kg (S1):** The root mass behaves violently due to lack of inertia. It acts as a "whiplash" source, showing the highest displacement in the chain (≈ 3.5 cm) and driving high-frequency jitters into the system.
- **At 1.0 kg (S2):** The system achieves an impedance match where wave propagation is efficient and the peak amplitudes are similar (≈ 2.1 – 2.6 cm across the masses).
- **At 2.0 kg (S3):** The root mass becomes an "anchor." It moves *less* than the downstream masses (1.8 cm vs 2.1 cm), effectively filtering the shock before it propagates.

3. Design Conclusion:

- Use **Low Mass (S1)** when the goal is **maximum excitation** (e.g., shakers, hammer drills).
- Use **High Mass (S3)** when the goal is **shock isolation** (e.g., heavy machine bases, armor plates).

3-DOF IMPULSE RESPONSE: MIDDLE MASS SENSITIVITY (m_2)

Subject: Full Spectrum Analysis of Impact Point Inertia ($m_2 = 0.5, 1.0, 2.0$ kg)

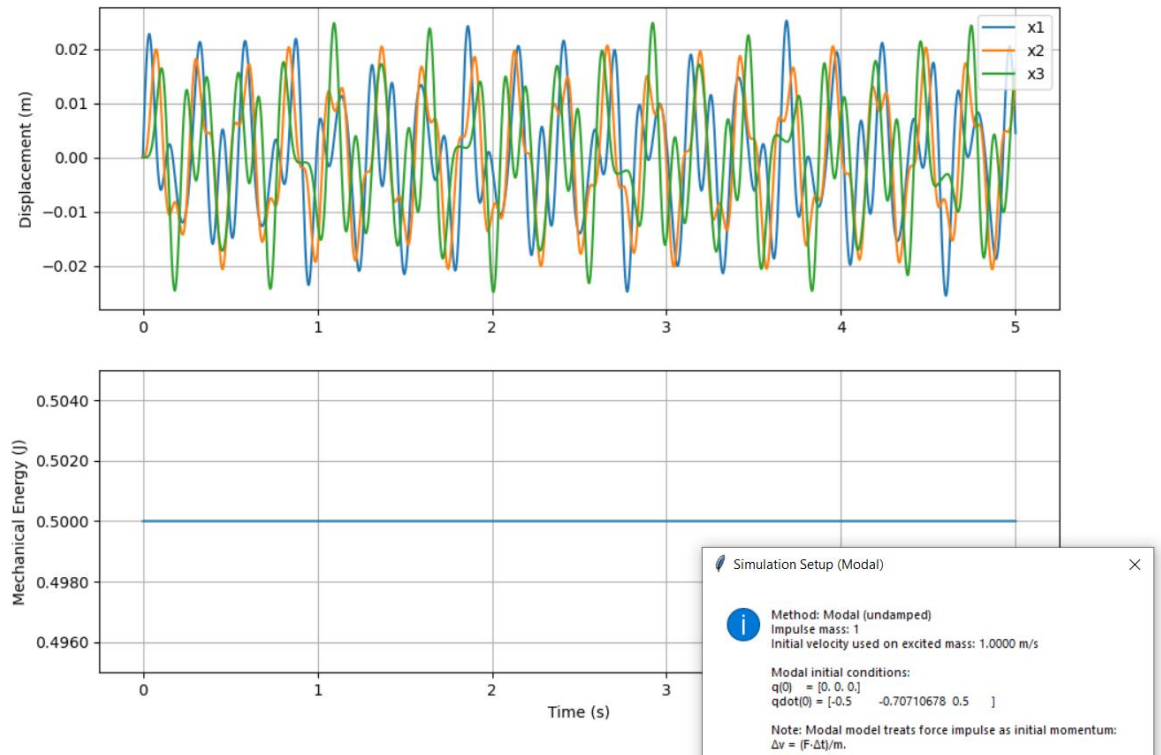


Figure 4: For $m_2 = 1.0$ kg

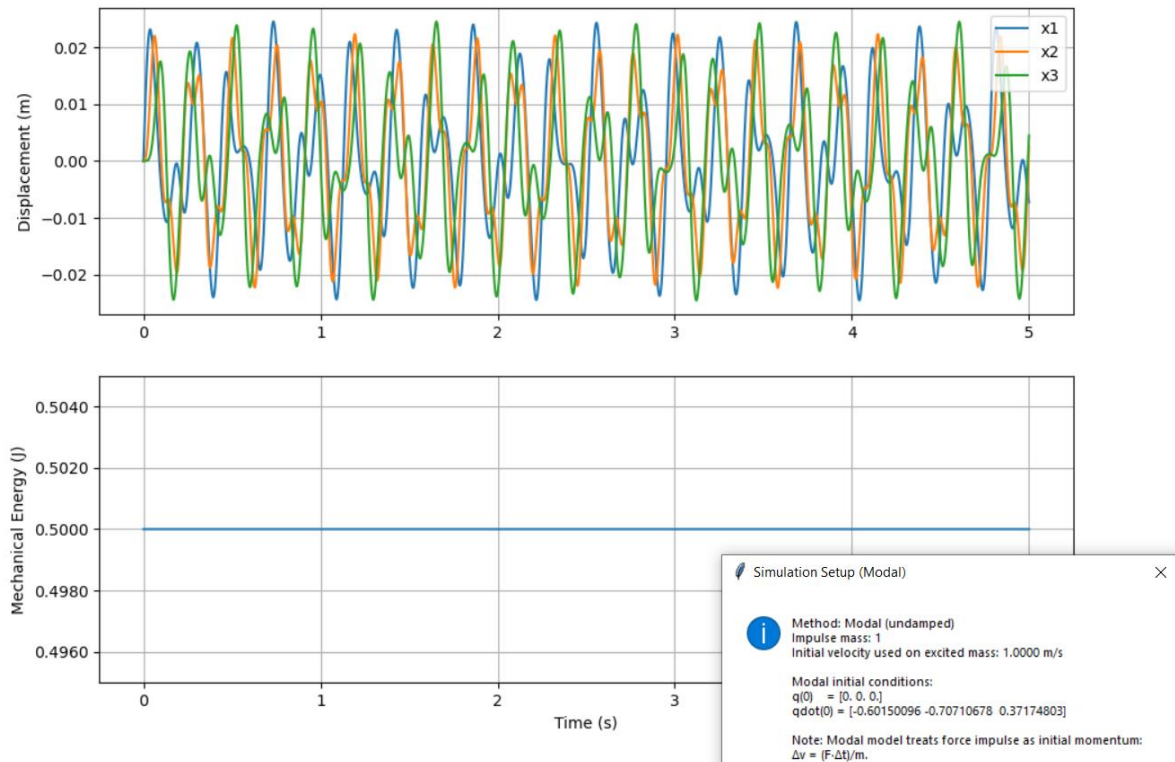


Figure 5: For $m_2 = 0.5$ kg

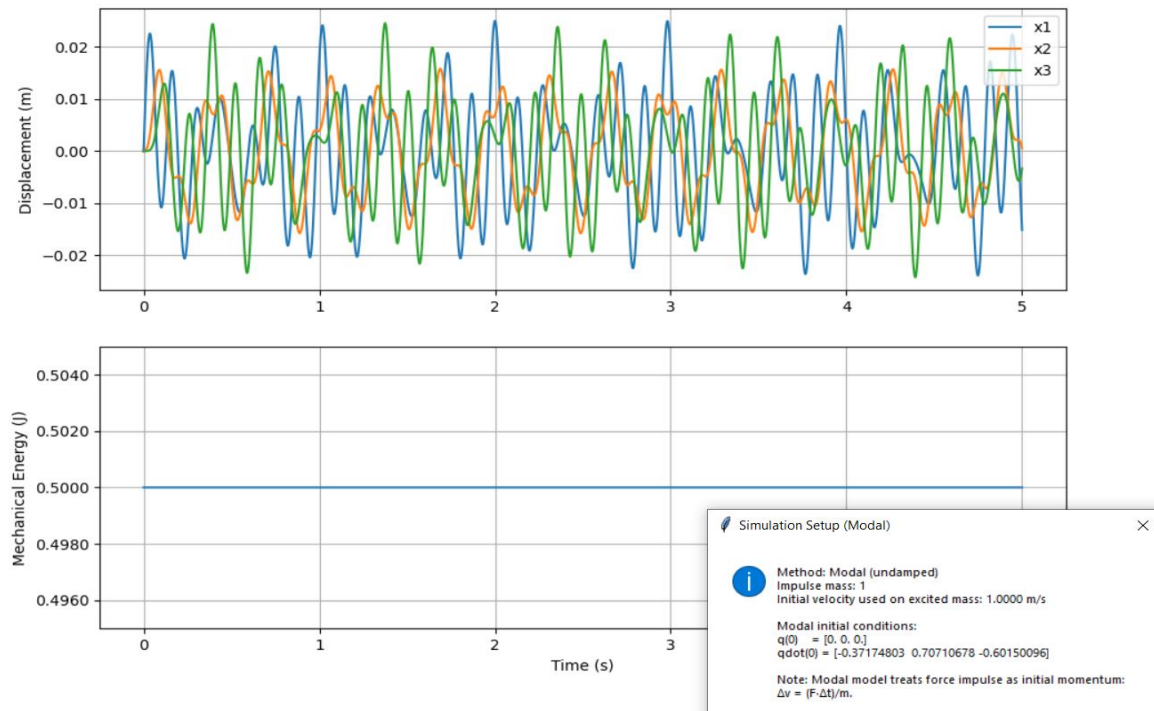


Figure 6: For $m_2 = 2.0 \text{ kg}$

COMPARATIVE TABLE 2

Feature	State S4 (0.5 kg)	State S5 (1.0 kg)	State S6 (2.0 kg)	Engineering Impact
Inertia Factor	0.5x (Low Impedance)	1.0x (Base)	2.0x (High Impedance)	Determines wave transmission.
Input Energy	0.50 J	0.50 J	0.50 J	Invariant (Source m_1 is constant).
Initial Velocity	1.0 m/s	1.0 m/s	1.0 m/s	Constant (v_0 depends on m_1)
Disp. Profile	Uniform ($x_1 \approx x_2 \approx x_3$)	Uniform ($x_1 \approx x_3$)	Notch ($x_2 \ll x_1, x_3$)	Shift from "Coupler" to "Barrier".
Response	Transmissive / Fluid	Nominal	Reflective / Isolating	Impedance Mismatch Effect.

Comparative Analysis (The "Mass Effect" Spectrum):

This 3-point analysis reveals the fundamental physics of wave propagation through the chain:

1. Energy Trend (Invariance):

Unlike the Mass 1 study, the Input Energy and Initial Velocity remain **constant** across all states. This confirms that changing downstream components (m_2) does not affect the energy admission at the source, but strictly controls how that energy is distributed.

2. Mode Shape Transformation:

- **At 0.5 kg (S4):** The middle mass has low inertia and acts as a "stiff coupler." It moves in perfect sync with the neighbors, allowing the wave to travel without loss.
- **At 1.0 kg (S5):** The system is balanced; wave propagation is nominal.
- **At 2.0 kg (S6):** The middle mass becomes an "inertial barrier." It moves significantly less ($\approx 1.5\text{cm}$) than the surrounding masses ($\approx 2.5\text{cm}$). This creates a "vibration shadow," reflecting energy back to the source (m_1) and isolating the tip.

3. Design Conclusion:

- Use **Low Mass (S4)** for **Power Transmission** (e.g., drive shafts, linkages) to ensure output equals input.
- Use **High Mass (S6)** for **Vibration Isolation** (e.g., heavy mounting blocks) to stop vibrations from reaching the tip.

3-DOF IMPULSE RESPONSE: TIP MASS SENSITIVITY (m_3)

Subject: Full Spectrum Analysis of Impact Point Inertia ($m_3 = 0.5, 1.0, 2.0$ kg)

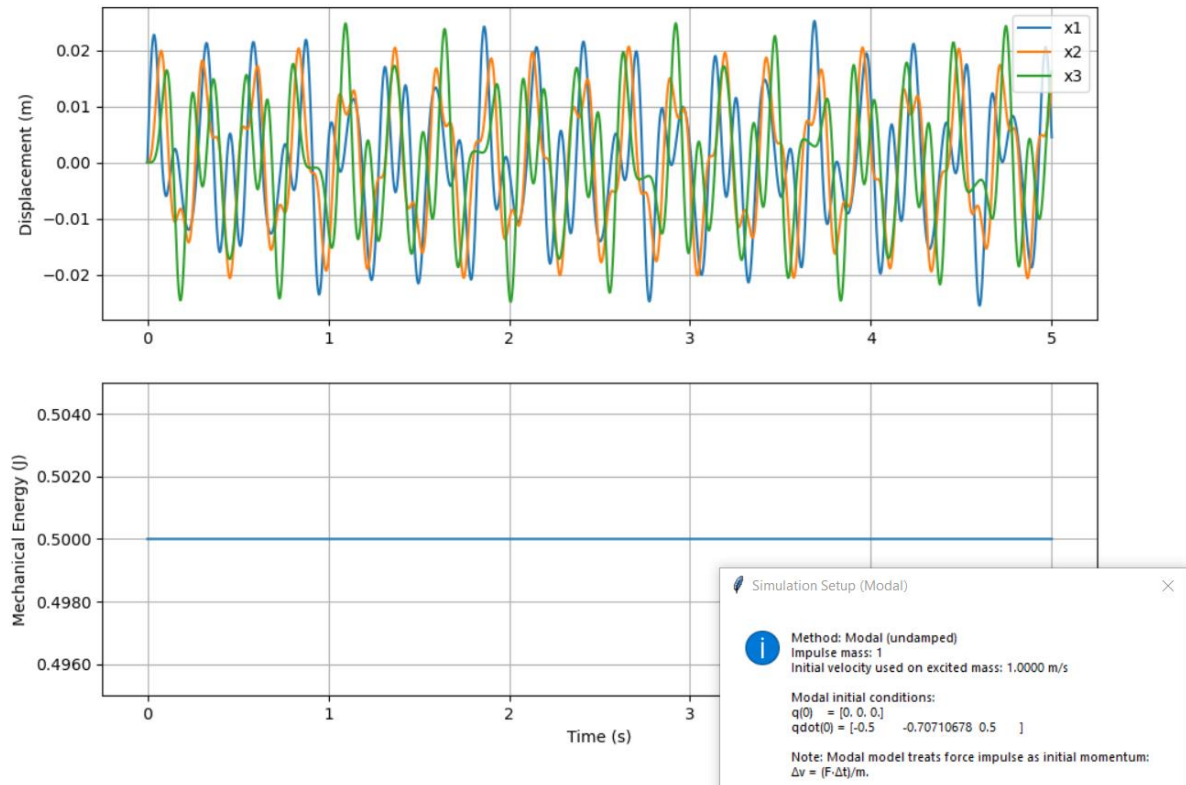


Figure 7: For $m_3 = 1.0$ kg

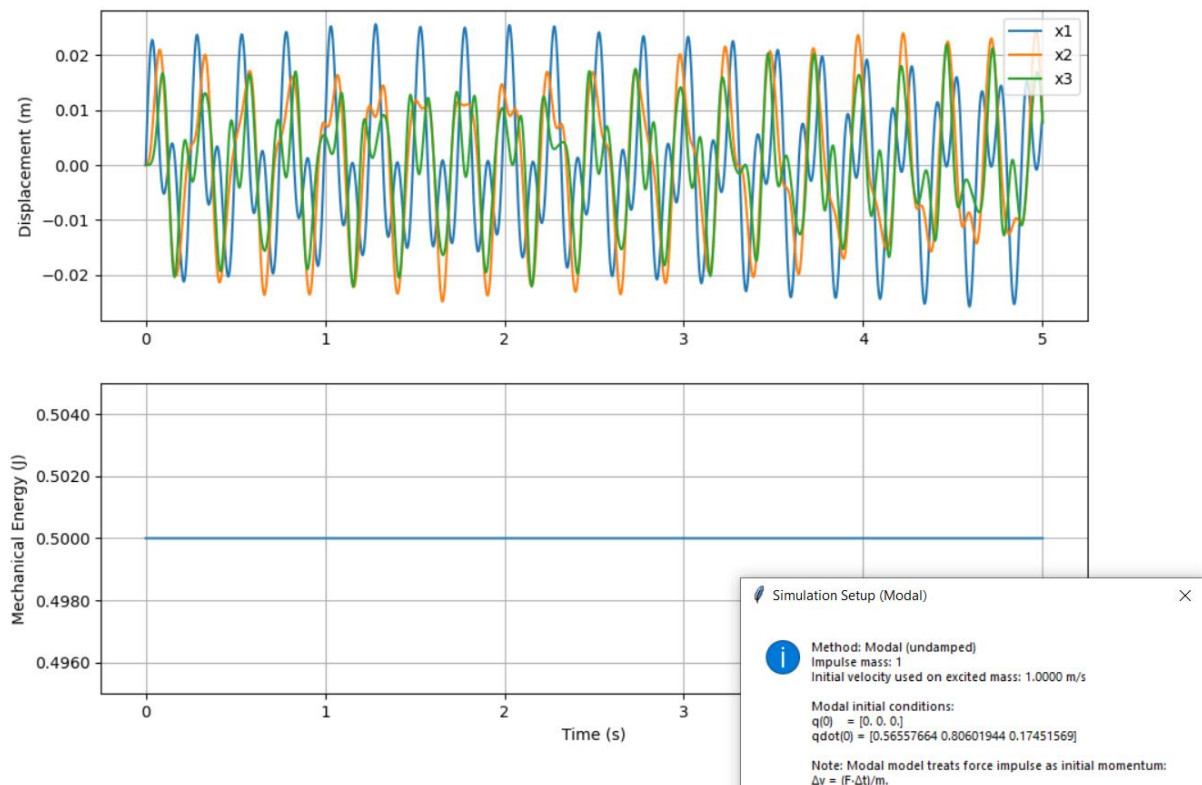


Figure 8: For $m_3 = 0.5$ kg

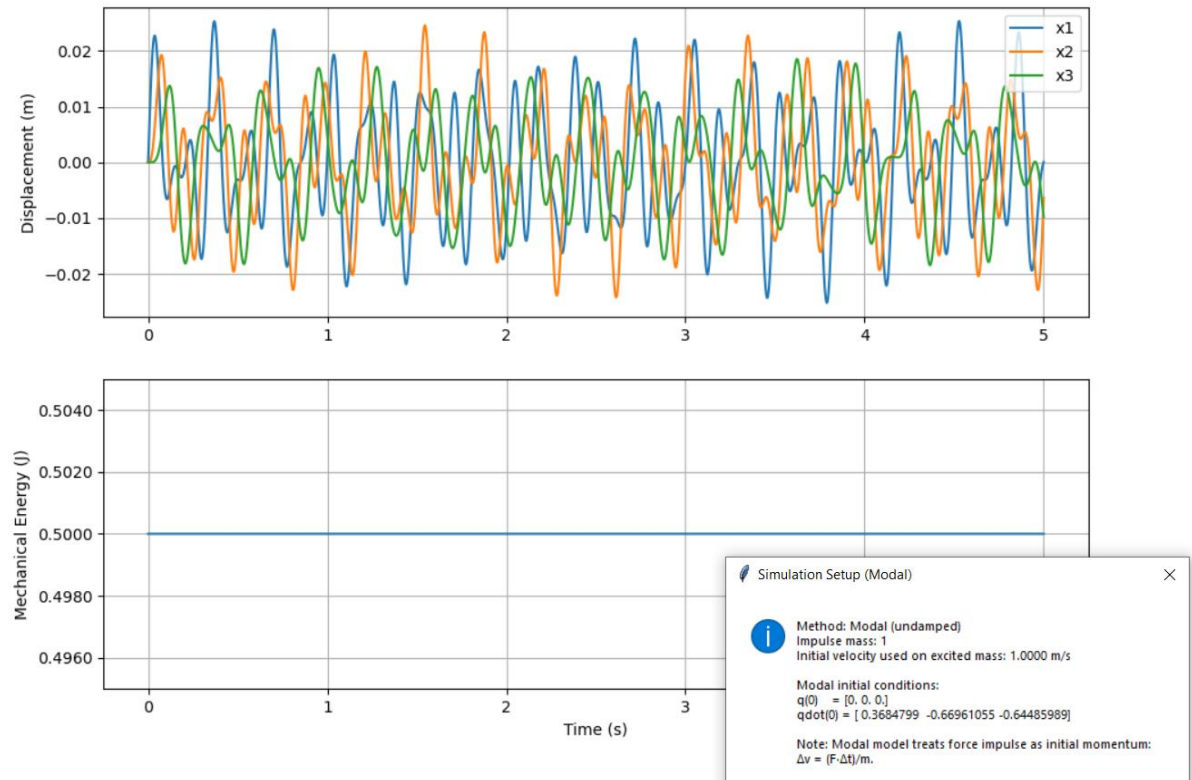


Figure 9: For $m_3 = 2.0 \text{ kg}$

COMPARATIVE TABLE 3

Feature	State S7 (0.5 kg)	State S8 (1.0 kg)	State S9 (2.0 kg)	Engineering Impact
Tip Inertia	0.5x (Low Impedance)	1.0x (Base)	2.0x (High Impedance)	Controls boundary condition.
Input Energy	0.50 J	0.50 J	0.50 J	Invariant (Source m_1 is constant).
Initial Velocity	1.0 m/s	1.0 m/s	1.0 m/s	Constant (v_0 depends on m_1).
Disp. Profile	Tip-Reduced ($x_3 < x_1$)	Uniform ($x_1 \approx x_3$)	Tip-Suppressed ($x_3 \ll x_1$)	"Whipping" vs. "Clamping".

Feature	State S7 (0.5 kg)	State S8 (1.0 kg)	State S9 (2.0 kg)	Engineering Impact
Response	Free-End Behavior	Nominal	Fixed-End Behavior	Wave reflection characteristics.

Comparative Analysis (The "Whipping vs. Anchor" Effect):

This final mass study demonstrates how the inertia at the end of the chain dictates how the traveling wave behaves when it reaches the termination point:

1. Energy Trend (Invariance):

Just like the Middle Mass study, changing the tip mass (m_3) has **zero effect** on the energy entering the system at m_1 . The total system energy remains constant at 0.50 J.

2. Mode Shape Transformation:

- **At 0.5 kg (S7):** The tip mass is light and offers little resistance to the incoming wave, but no **"Whipping Effect"** appears: the light tip has the smallest peak displacement in the chain (≈ 2.2 cm, versus ≈ 2.6 cm at the root mass x_1).
- **At 1.0 kg (S8):** The system is balanced, showing uniform distribution.
- **At 2.0 kg (S9):** The tip mass acts as a "soft wall" or **Anchor**. It refuses to move (≈ 1.9 cm) relative to the rest of the chain. Interestingly, this causes the energy to "pile up" in the middle mass (x_2), raising its peak from ≈ 2.1 cm to ≈ 2.5 cm as the wave reflects back rather than dissipating into tip motion.

3. Design Conclusion:

- **Low Mass (S7)** does not amplify the tip motion in this model, so it is not a way to boost sensor signals at the tip.
- Use **Heavy Mass (S9)** for **Stabilizers** (e.g., a counterweight) to minimize motion at the end of a structure, effectively creating a pseudo-fixed boundary.

SECTION 2 — SPRING STIFFNESS VARIATION (k_0)

Subject: Analysis of Wall Stiffness / Root Constraint ($k_0 = 500, 1000, 2000 \text{ N/m}$)

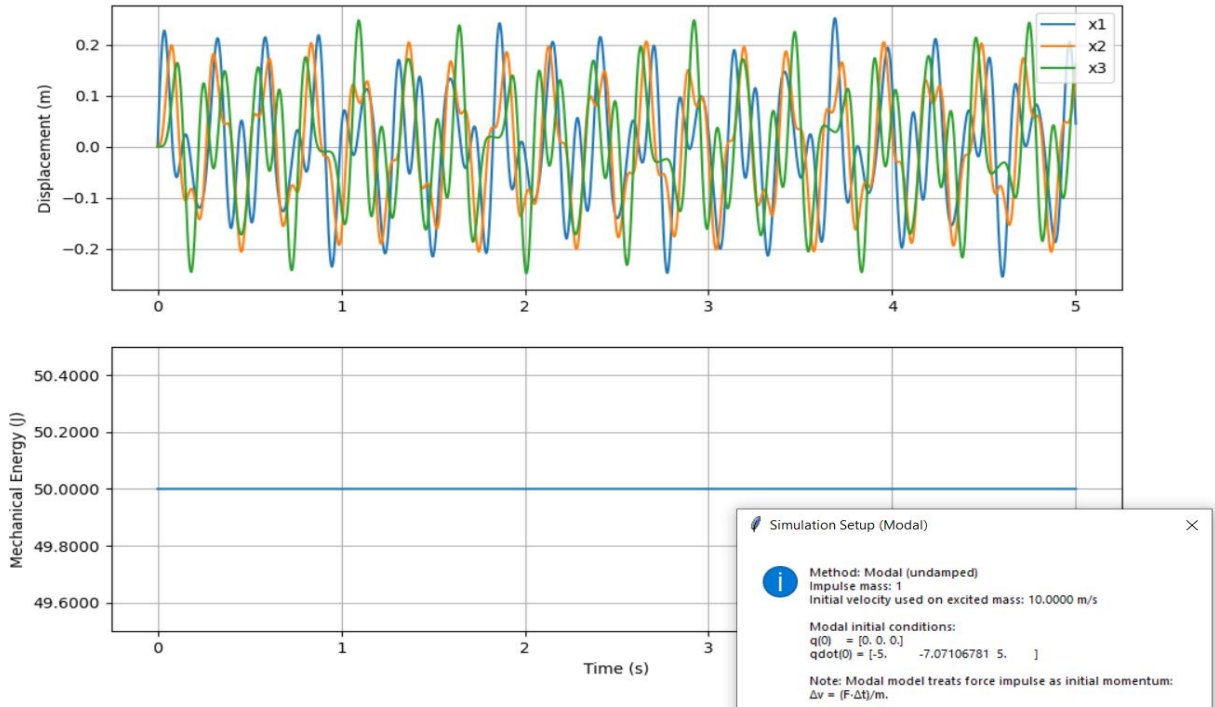


Figure 10: For $k_0 = 1000 \text{ N/m}$

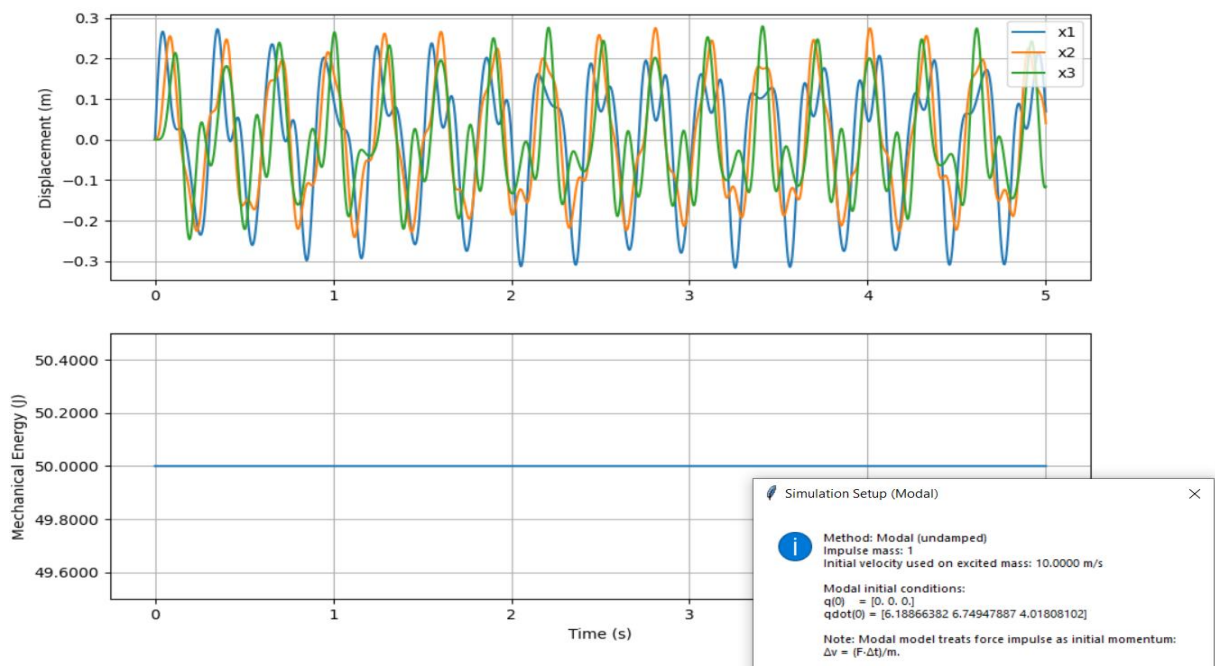


Figure 11: For $k_0 = 500 \text{ N/m}$

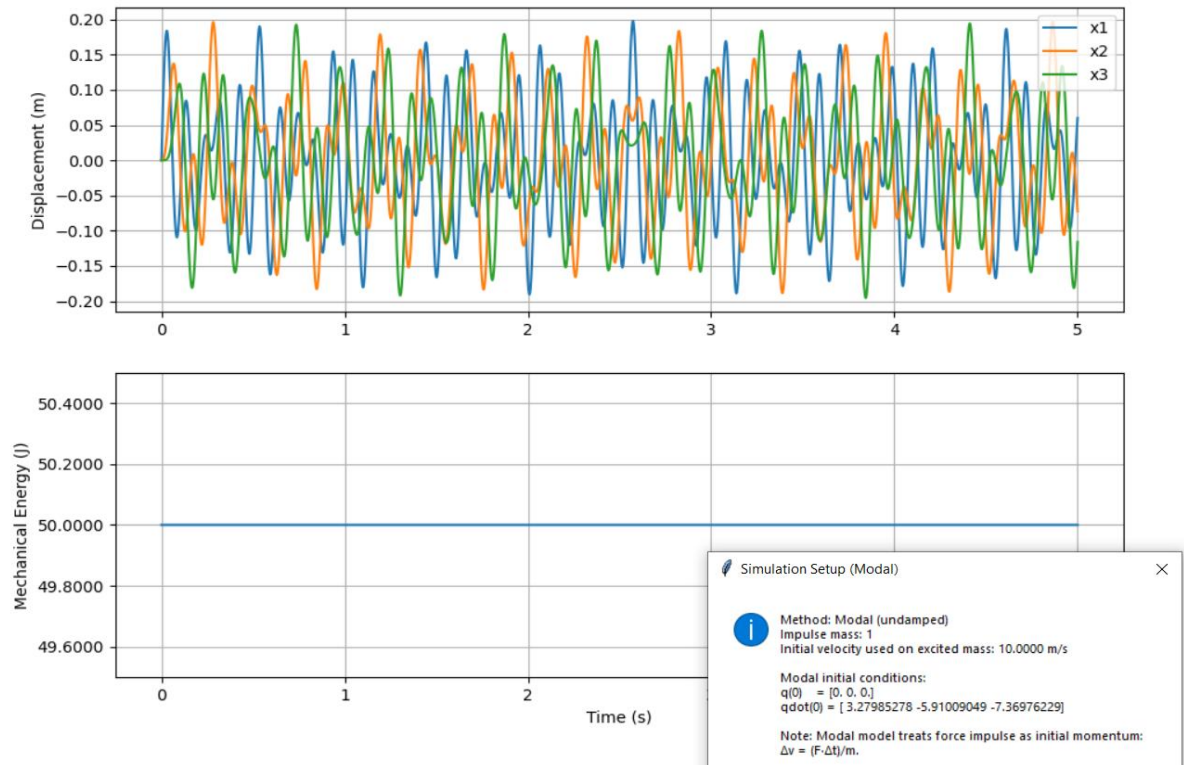


Figure 12: For $k_0 = 2000 \text{ N/m}$

COMPARATIVE TABLE 4

Feature	State S10 (Soft - 500)	State S0 (Ref - 1000)	State S11 (Stiff - 2000)	Engineering Impact
Root Stiffness	0.5x (Flexible)	1.0x (Base)	2.0x (Rigid)	Defines system constraint.
Input Energy	50.0 J	50.0 J	50.0 J	Invariant (E depends on mass, not spring).
Response Speed	Slow / Long Period	Nominal	Fast / Short Period	Frequency $w_n \propto \sqrt{k}$
Root Disp. (x_1)	High ($\approx 0.32 \text{ m}$)	Nominal ($\approx 0.26 \text{ m}$)	Low ($\approx 0.20 \text{ m}$)	Stiff spring limits travel.
System Role	Suspension / Isolator	General Structure	Rigid Mount	Precision vs. Comfort.

Comparative Analysis (The "Stiffness Control" Spectrum):

This analysis isolates the effect of the anchoring spring (k_0) on the system dynamics:

1. Energy Trend (Invariance):

Unlike the Mass variation, changing the spring stiffness **does not change the energy** admitted into the system (remains constant at 50.0 J). Impulse energy is purely kinetic initially ($E = \frac{I^2}{2m}$), which is independent of stiffness.

2. Displacement & Frequency Shift:

- **At 500 N/m (S10):** The system is "soft." The root mass (m_1) experiences large excursions (≈ 32 cm) because the restoring force is weak. The oscillation period is long (slow waves).
- **At 2000 N/m (S11):** The system is "stiff." The root mass is tightly constrained to the wall, limiting its movement to just ≈ 20 cm. However, this causes the vibration frequency to increase, resulting in "snappier," faster oscillations.

3. Design Conclusion:

- Use **Low Stiffness (S10)** for **Shock Absorption** (e.g., vehicle suspension) where large travel is acceptable to lower the transmitted acceleration.
- Use **High Stiffness (S11)** for **Precision Positioning** (e.g., CNC machine mounts) where displacement must be minimized despite high loads.

SPRING STIFFNESS VARIATION (k_1)

Subject: Analysis of Coupling Stiffness ($k_1 = 500, 1000, 2000 \text{ N/m}$)

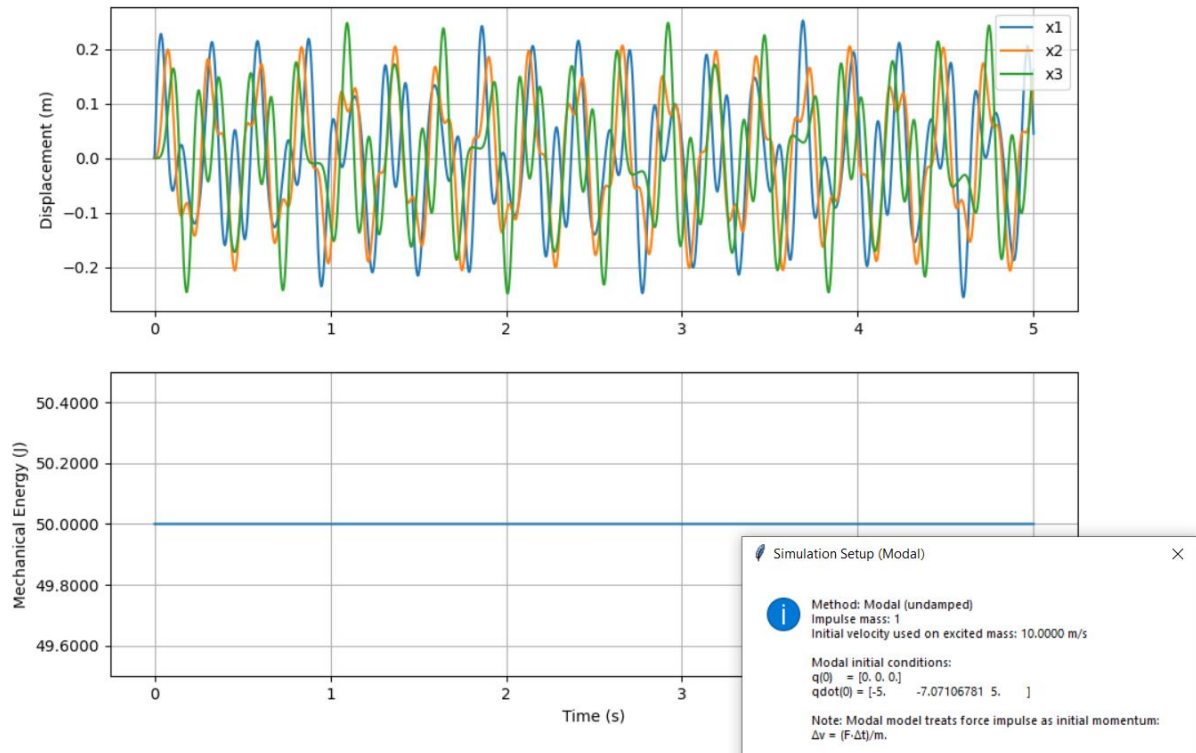


Figure 13: $k_1 = 1000 \text{ N/m}$

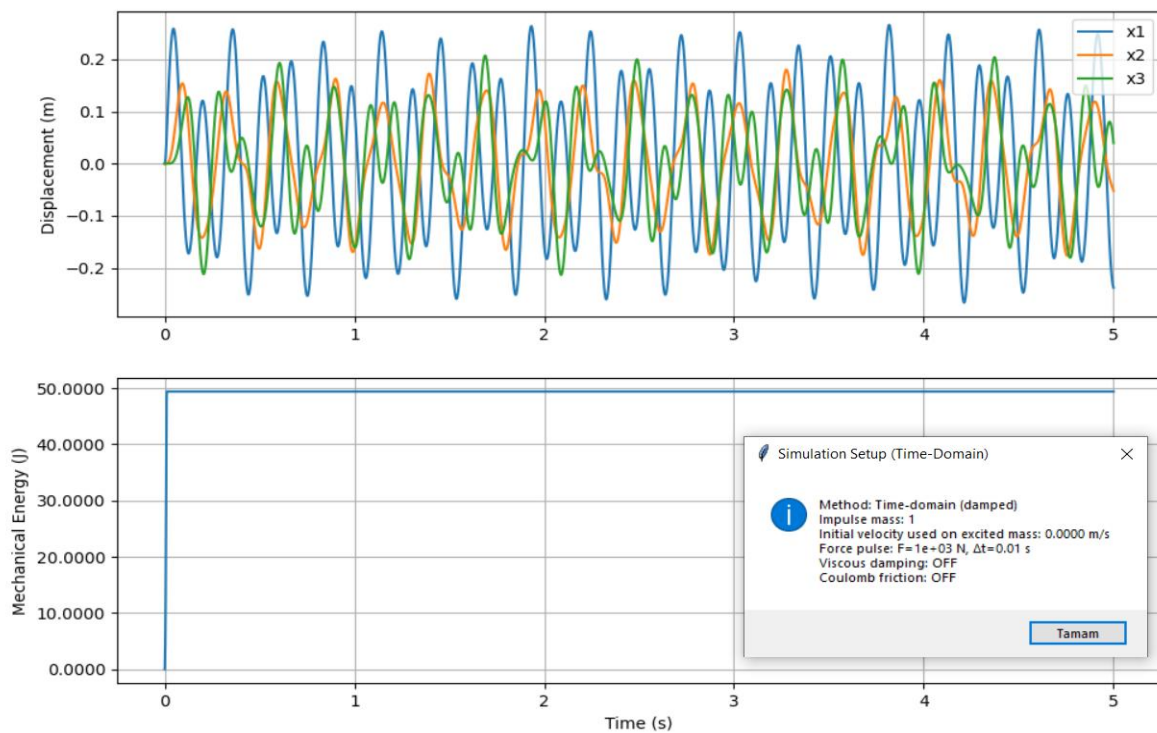


Figure 14: For $k_1 = 500 \text{ N/m}$

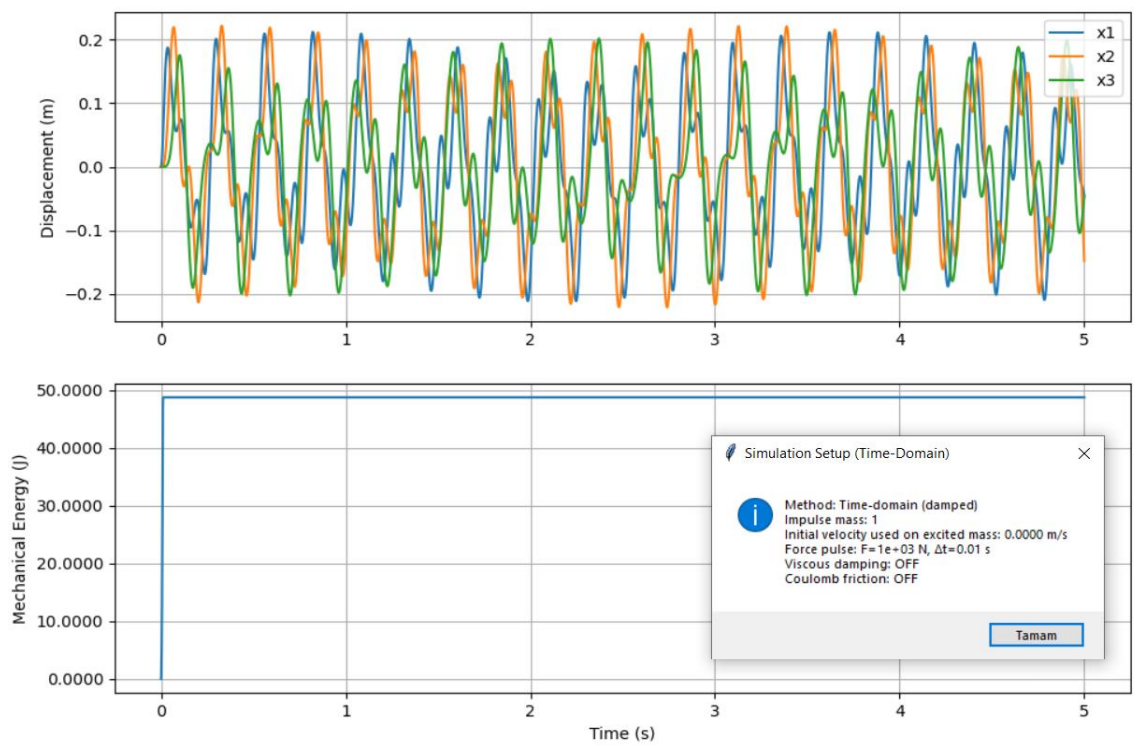


Figure 15: For $k_1 = 2000 \text{ N/m}$

COMPARATIVE TABLE 5

Feature	State S36 (Soft - 500)	State S0 (Ref - 1000)	State S37 (Stiff - 2000)	Engineering Impact
Coupling Strength	0.5x (Weak)	1.0x (Base)	2.0x (Strong)	Transmission efficiency.
Input Energy	50.0 J	50.0 J	50.0 J	Invariant (E is kinetic).
Energy Transfer	Delayed / Lagging	Nominal	Instant / Direct	Response time to impulse.
Source Disp. (x_1)	High (Loose)	Nominal	Low (Coupled)	Interaction with wall spring k_0 .
System Role	Vibration Filter	Structural Link	Rigid Drive	Isolation vs. Actuation.

Comparative Analysis (The "Transmission Stiffness" Spectrum):

This study isolates the effect of the spring connecting the excited mass (m_1) to the rest of the chain (m_2):

1. Energy Trend (Invariance):

Consistent with previous spring tests, the total energy remains fixed at 50.0 J. Stiffness alters the *flow* of energy, not the *amount*.

2. Coupling Dynamics:

- **At 500 N/m (S36):** The connection is "soft." Mass 1 can move significantly before Mass 2 feels the pull. This creates a **phase lag** and acts as a mechanical filter, smoothing out sharp jerks before they reach the rest of the chain. The waveform shows distinct "beating" (energy slopping back and forth slowly).
- **At 2000 N/m (S37):** The connection is "stiff." Mass 1 and Mass 2 are tightly bonded. When m_1 moves, m_2 is forced to follow almost immediately. This increases the system's effective natural frequency and allows high-frequency shock content to pass through to the tail (m_3).

3. Design Conclusion:

- Use **Low Stiffness (S36)** for **Isolation Couplings** (e.g., flexible motor couplings) to prevent source vibrations from propagating down the line.
- Use **High Stiffness (S37)** for **Drive Trains** where immediate response and precise motion transfer are required (e.g., servo mechanisms).

SPRING STIFFNESS VARIATION (k_2)

Subject: Analysis of Coupling Stiffness ($k_2 = 500, 1000, 2000 \text{ N/m}$)

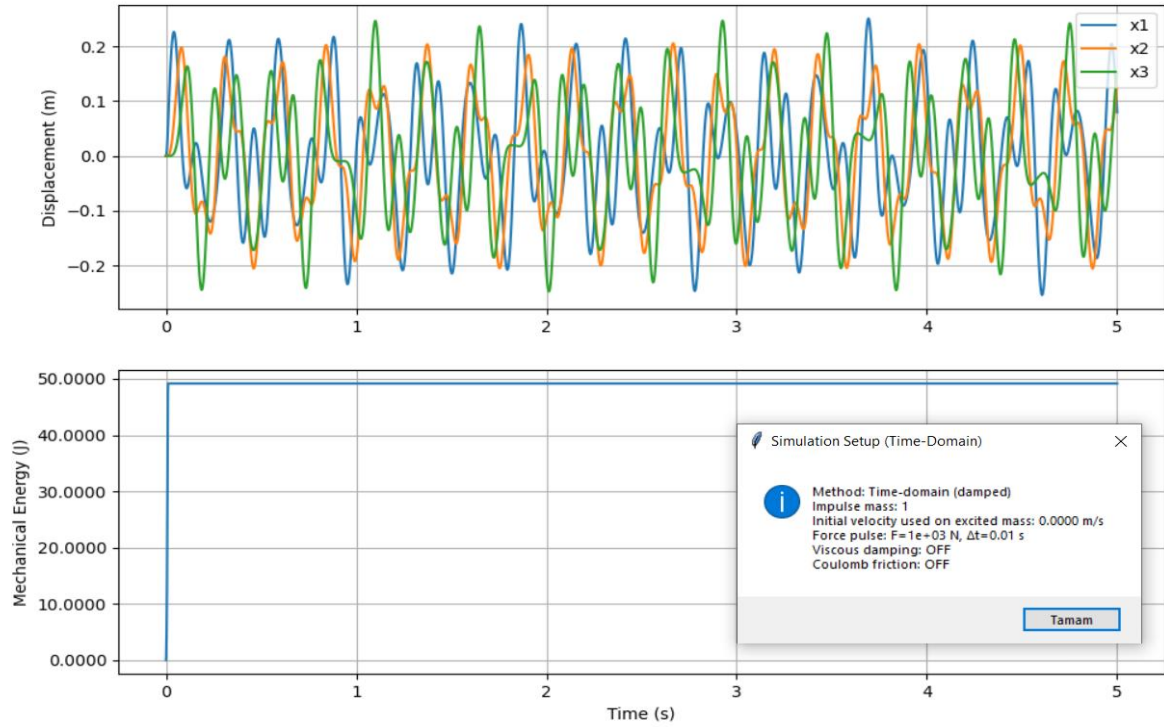


Figure 16: For $k_2 = 1000 \text{ N/m}$

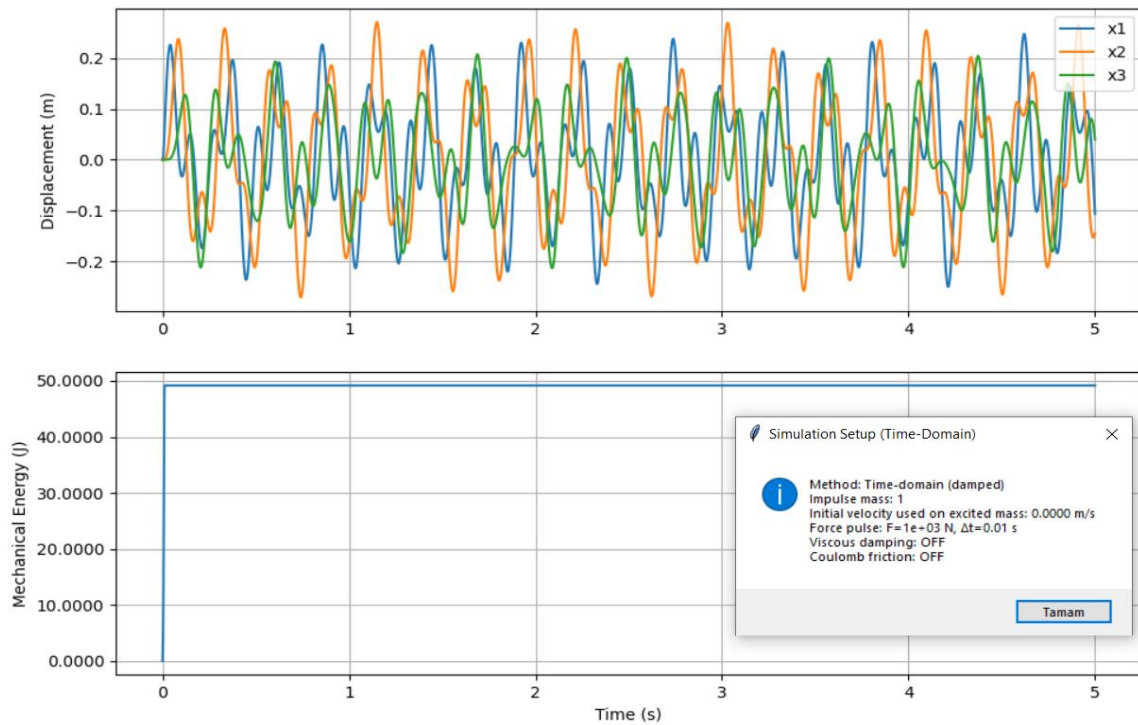


Figure 17: For $k_2 = 500 \text{ N/m}$

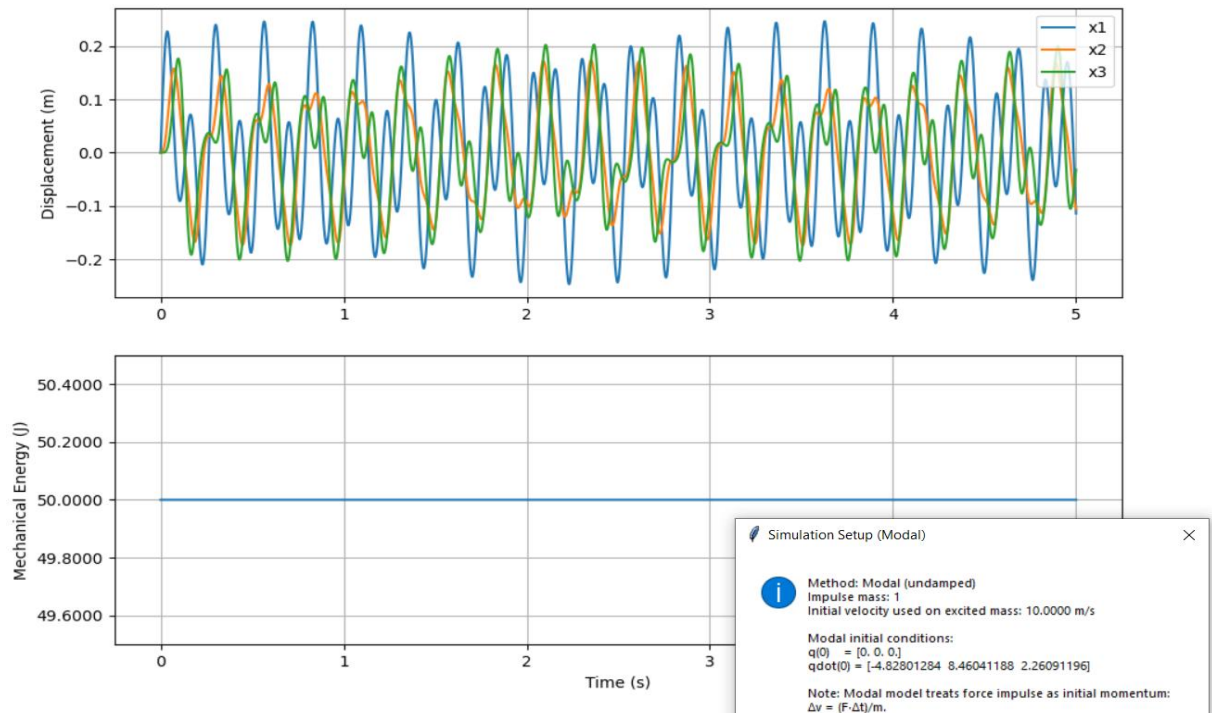


Figure 18: $For k_2 = 2000 \text{ N/m}$

COMPARATIVE TABLE 6

Feature	State S12 (Soft - 500)	State S0 (Ref - 1000)	State S13 (Stiff - 2000)	Engineering Impact
Tail Connection	0.5x (Loose)	1.0x (Base)	2.0x (Rigid)	Defines endpoint control.
Input Energy	~50.0 J	~50.0 J	~50.0 J	Invariant (Source is far from tail).
Tip Response (x_3)	Low / Delayed	Nominal	Low / Instant	Wave arrival timing.
Mode Shape	Localized Tail Mode	Distributed	Unified Motion ($x_2 \approx x_3$)	m_2 and m_3 lock together.
System Role	Flexible Appendage	Standard Link	Rigid Extension	Whip vs. Rod behavior.

Comparative Analysis (The "Tail Whip" Spectrum):

This study isolates the effect of the spring connecting the end-mass (m_3) to the chain:

1. Energy Trend (Invariance):

Changing the tail spring (k_2) has absolutely no effect on the energy admitted at the head (m_1). The system absorbs the same shock (~ 50 J), but dissipates/distributes it differently.

2. Coupling Dynamics:

- **At 500 N/m (S12):** The tail connection is loose. Mass 3 is only weakly driven: its peak drops to ≈ 0.21 m (from ≈ 0.25 m), while Mass 2 reaches the largest peak in the chain (≈ 0.27 m).
- **At 2000 N/m (S13):** The tail connection is stiff. Mass 2 and Mass 3 are effectively locked together. They move with similar amplitudes and phase, acting like a single heavier block. This reduces the independent vibration of the tip but increases the effective inertia seen by the rest of the chain.

3. Design Conclusion:

- Use **Low Stiffness (S12)** for **Flexible Booms** or antennas where compliance is needed to avoid structural stress at the joint.
- Use **High Stiffness (S13)** for **Robot Arms** or positioning stages where the tip must faithfully track the motion of the actuator without wobbling.

SPRING STIFFNESS VARIATION (k_3)

Subject: Analysis of Wall Stiffness / End Constraint ($k_3 = 500, 1000, 2000 \text{ N/m}$)

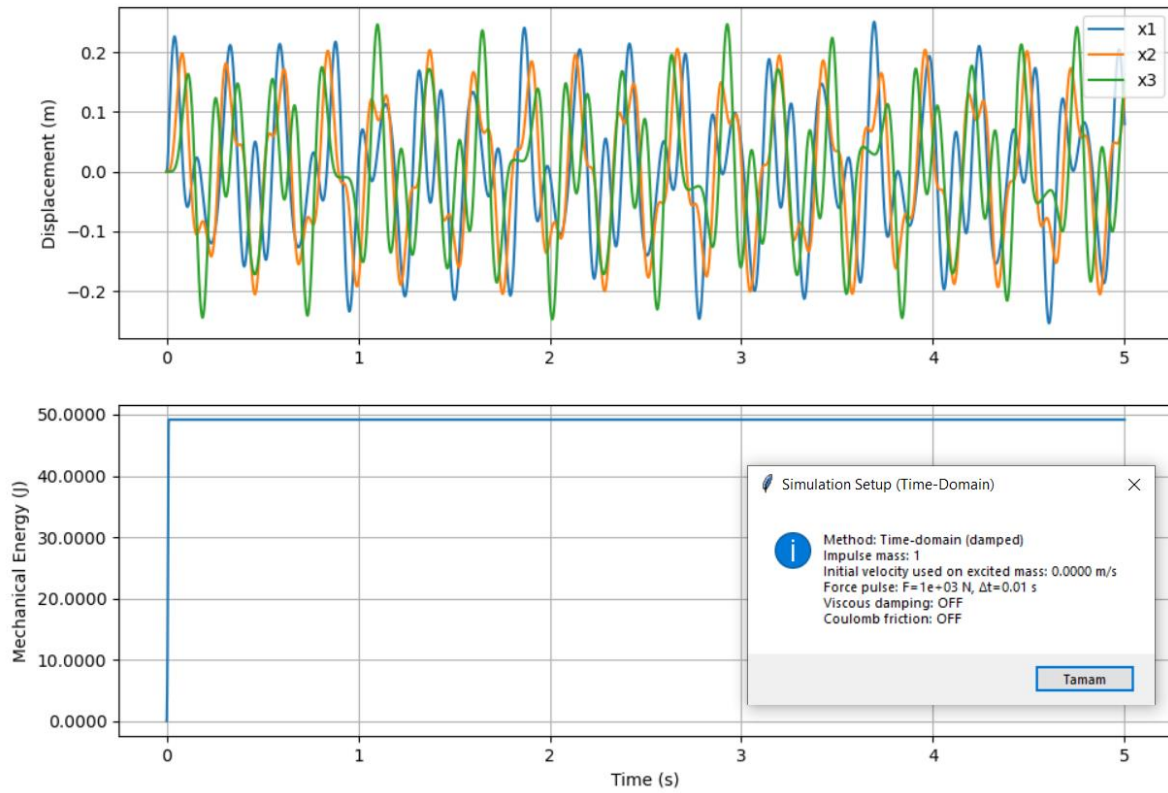


Figure 19: For $k_3 = 1000 \text{ N/m}$

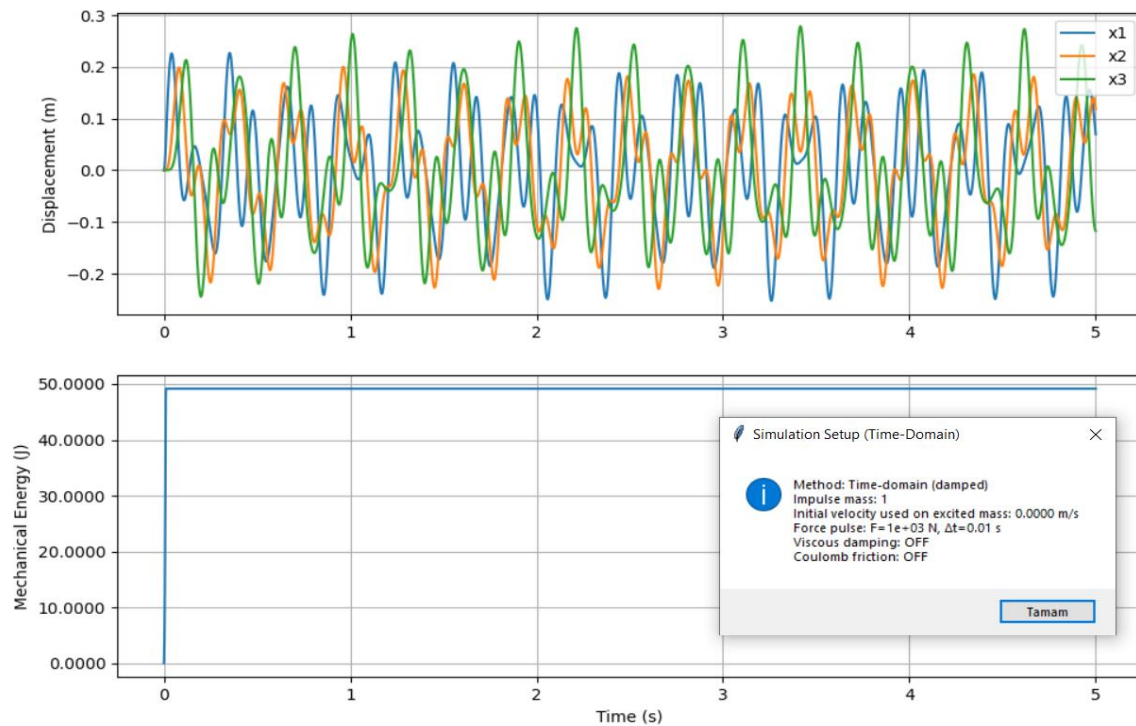


Figure 20: For $k_3 = 500 \text{ N/m}$

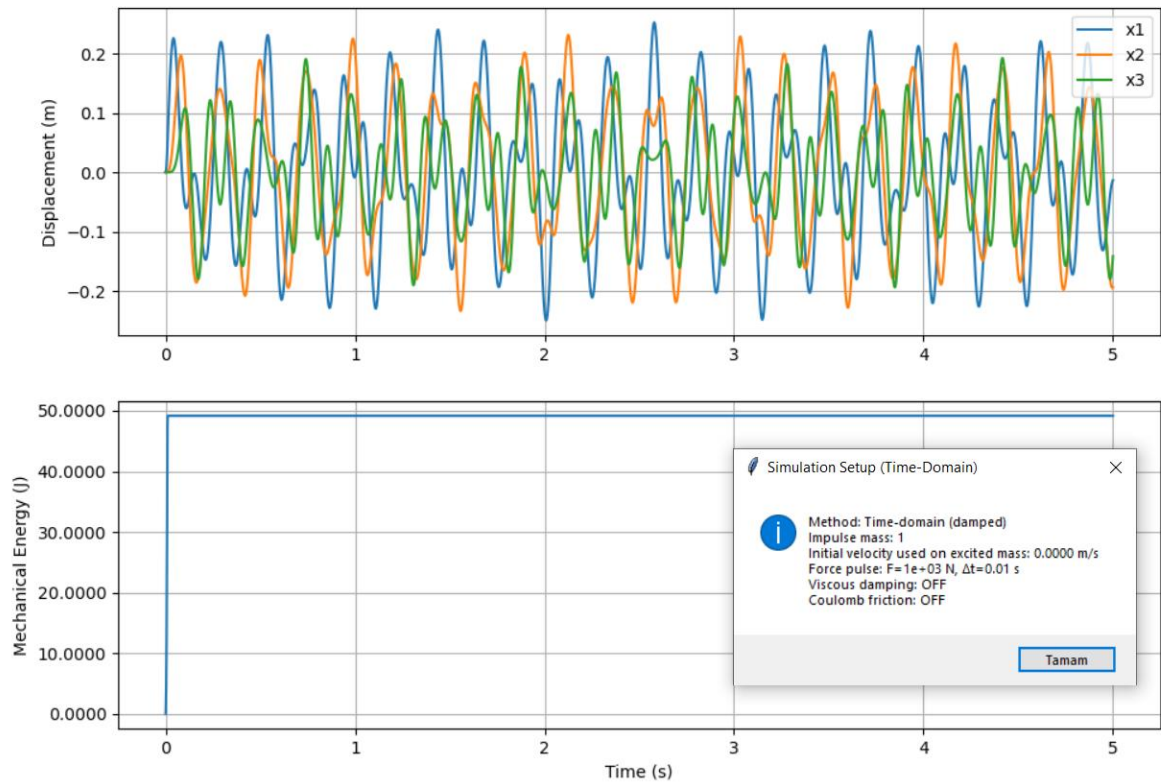


Figure 21: For $k_3 = 2000 \text{ N/m}$

COMPARATIVE TABLE 7

Feature	State S14 (Soft - 500)	State S0 (Ref - 1000)	State S15 (Stiff - 2000)	Engineering Impact
End Constraint	0.5x (Flexible)	1.0x (Base)	2.0x (Rigid)	Defines boundary condition.
Input Energy	~50.0 J	~50.0 J	~50.0 J	Invariant (Source is far from end).
Tip Disp. (x_3)	High (≈ 0.28 m)	Nominal	Low (≈ 0.20 m)	Clamped vs. Free behavior.
Reflection	Low (Absorbing)	Moderate	High (Reflecting)	Wave energy return.
System Role	Soft Catch / Buffer	Standard Anchor	Hard Wall	Structural termination type.

Comparative Analysis (The "Boundary Condition" Spectrum):

This study isolates the effect of the final anchoring spring (k_3) on the system dynamics:

1. Energy Trend (Invariance):

As with all spring tests, the stiffness of the final anchor has zero effect on the initial energy admitted at the start of the chain. The system processes the same ~ 50 J.

2. Constraint Dynamics:

- **At 500 N/m (S14):** The anchor is "soft." The system behaves somewhat like a fixed-free cantilever. Mass 3 swings with large amplitude (≈ 0.28 m), absorbing the wave energy into large potential energy excursions.
- **At 2000 N/m (S15):** The anchor is "stiff." The system behaves like a fixed-fixed beam. Mass 3 is strictly pinned down (≈ 0.20 m). Crucially, because the wave cannot move x_3 much, the energy is **reflected** back towards the source, while the peak displacement at x_1 remains practically the same as in the balanced state (0.254 m vs 0.255 m).

3. Design Conclusion:

- Use **Low Stiffness (S14)** for **End-Stop Buffers** (e.g., railway bumpers) where the goal is to decelerate the mass over a long distance.
- Use **High Stiffness (S15)** for **Structural Anchors** (e.g., bridge cables) where the end point must remain static under load.

SECTION 3 — SPRING LENGTH VARIATION (L_0)

Subject: Verification of Geometric Independence ($L_0 = 0.25, 0.5, 1.0$ m)

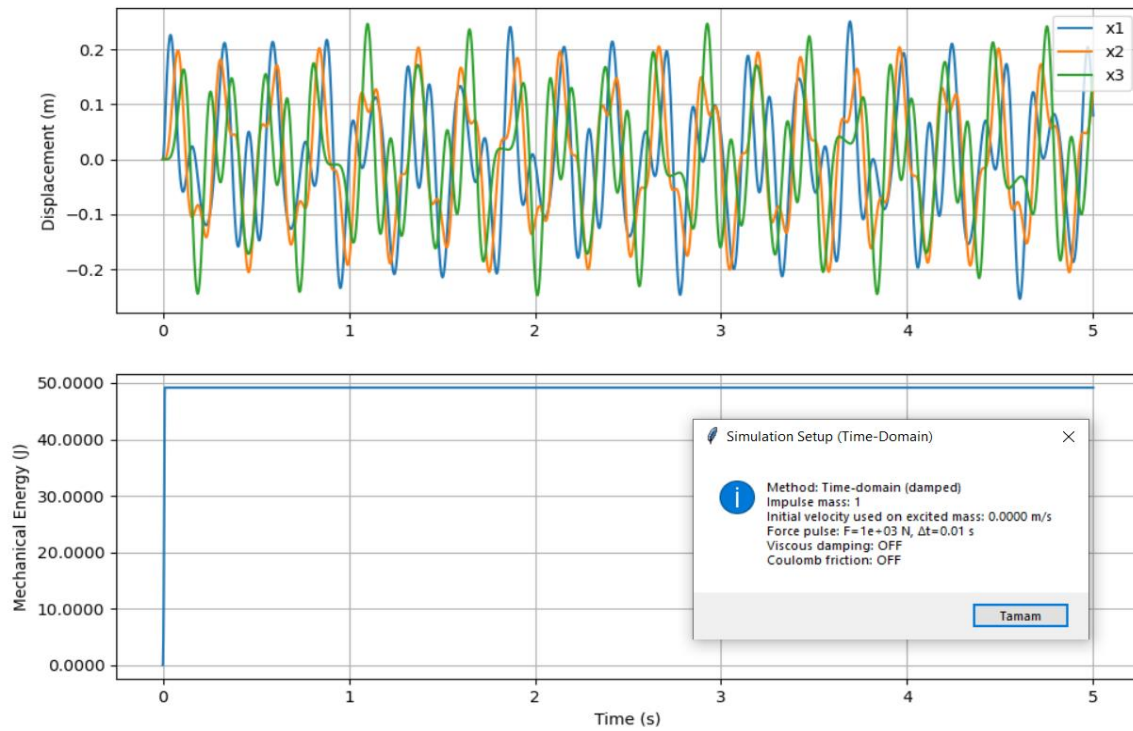


Figure 22: For $L_0 = 0.5$ m

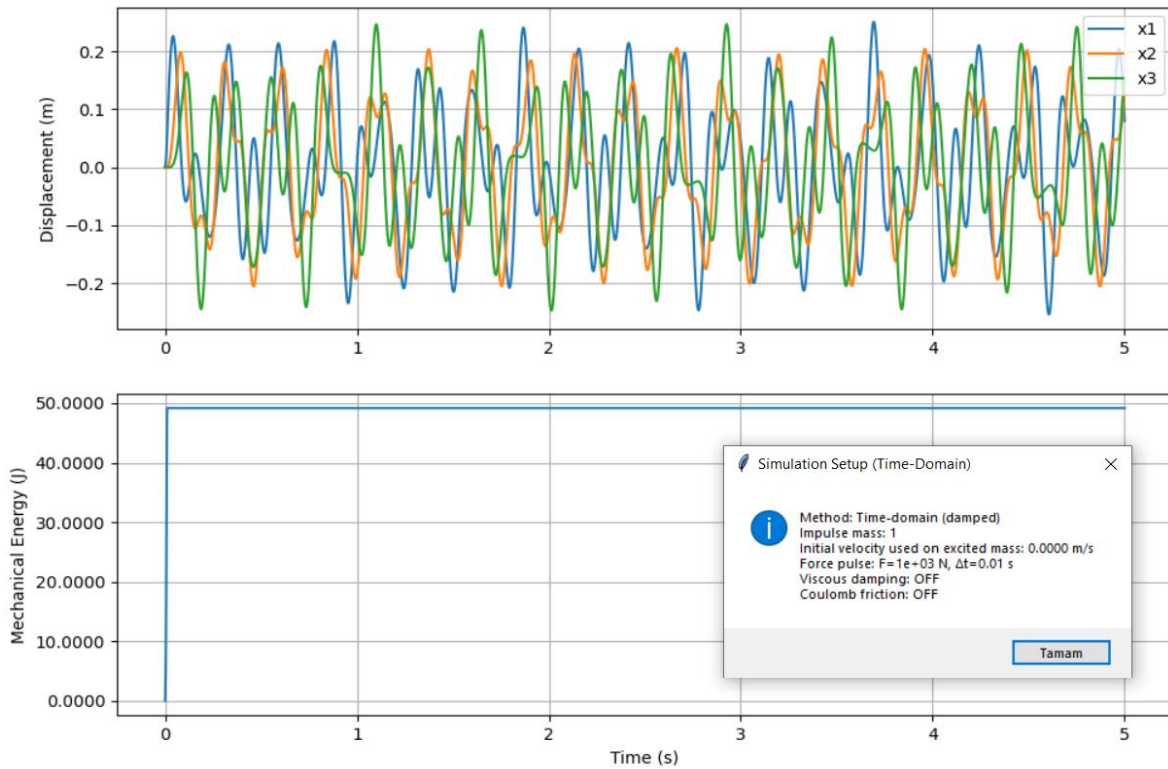


Figure 23: For $L_0 = 0.25$ m

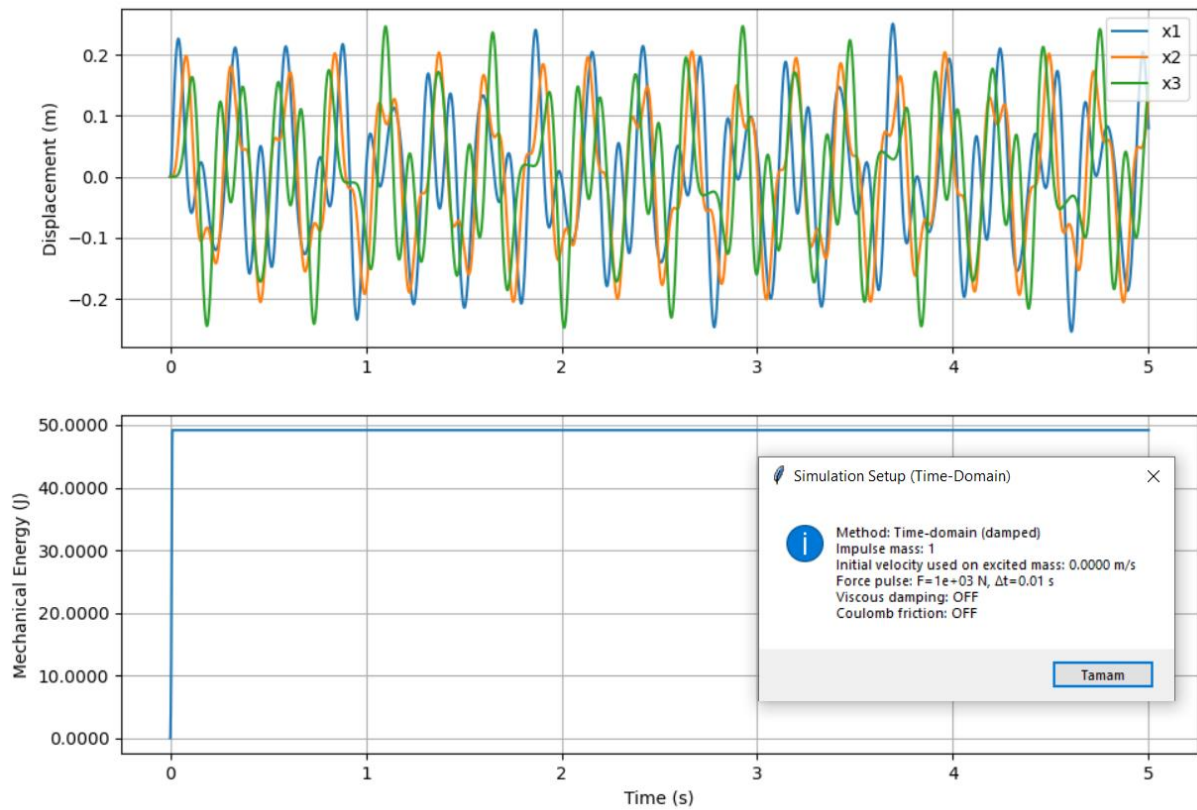


Figure 24: For $L_0 = 1$ m

COMPARATIVE TABLE 8

Feature	State S16 (Short)	State S0 (Ref)	State S17 (Long)	Engineering Impact
Geometric Length	0.5x (Compact)	1.0x (Base)	2.0x (Extended)	Visual spacing only.
Input Energy	50.0 J	50.0 J	50.0 J	Invariant
Frequencies	Constant	Constant	Constant	Dynamics are length-independent.
Stiffness (k)	Constant (1000)	Constant (1000)	Constant (1000)	k is an explicit input.
System Role	Lumped Model	Lumped Model	Lumped Model	Topography validation.

Comparative Analysis (Geometric Invariance Verification):

This test series serves as a Model Validation step to confirm that the simulation operates as a linear "Lumped Parameter" system ($F = -kx$).

1. Dynamic Invariance:

Changing the rest length of the springs (L_0) resulted in **zero change** in the displacement time-histories or energy levels. This confirms that the stiffness k is defined independently of the length (unlike a cantilever beam where $k \propto 1/L^3$).

2. Physical Interpretation:

In this mathematical model, the springs are idealized force elements. The "length" parameter strictly defines the equilibrium coordinates for visualization but does not introduce geometric stiffness or pendulum effects.

3. Design Conclusion:

For this specific system model, physical packaging (compact vs. long) does not alter the vibration isolation performance, provided the **spring rate (k)** remains constant.

SPRING LENGTH VARIATION (L_1)

Subject: Verification of Geometric Independence ($L_1 = 0.5, 1.0, 2.0$ m)

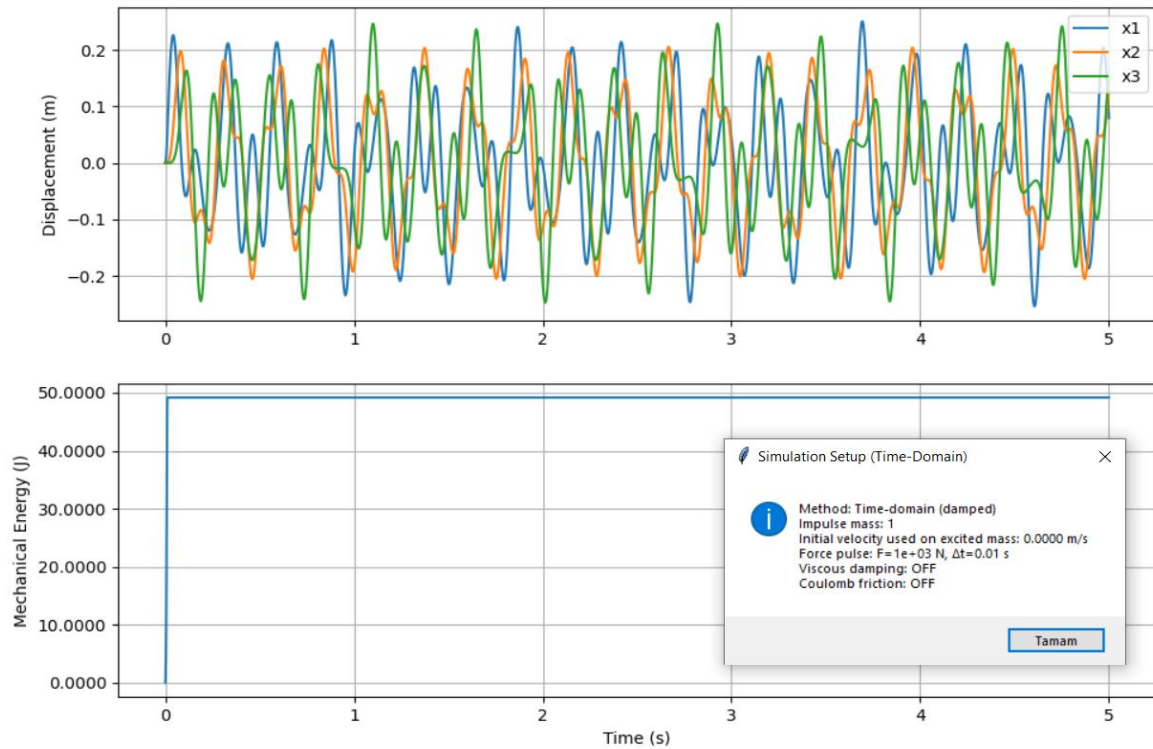


Figure 25: For $L_1 = 1.0$ m

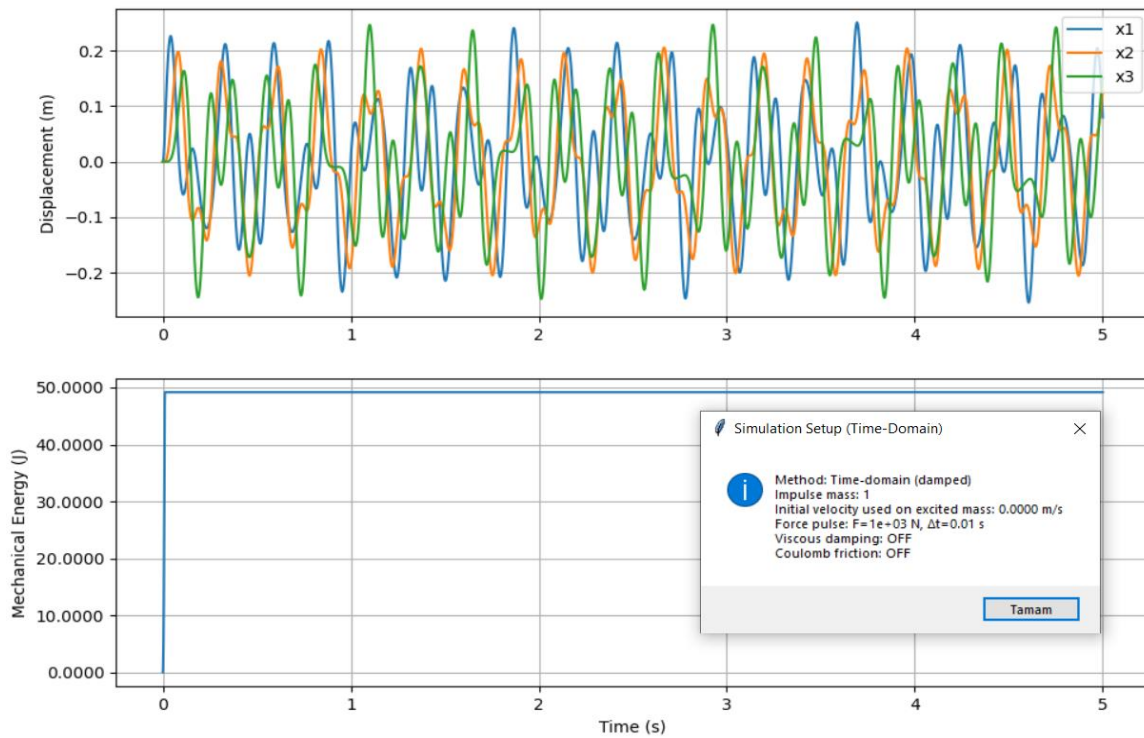


Figure 26: For $L_1 = 0.5$ m

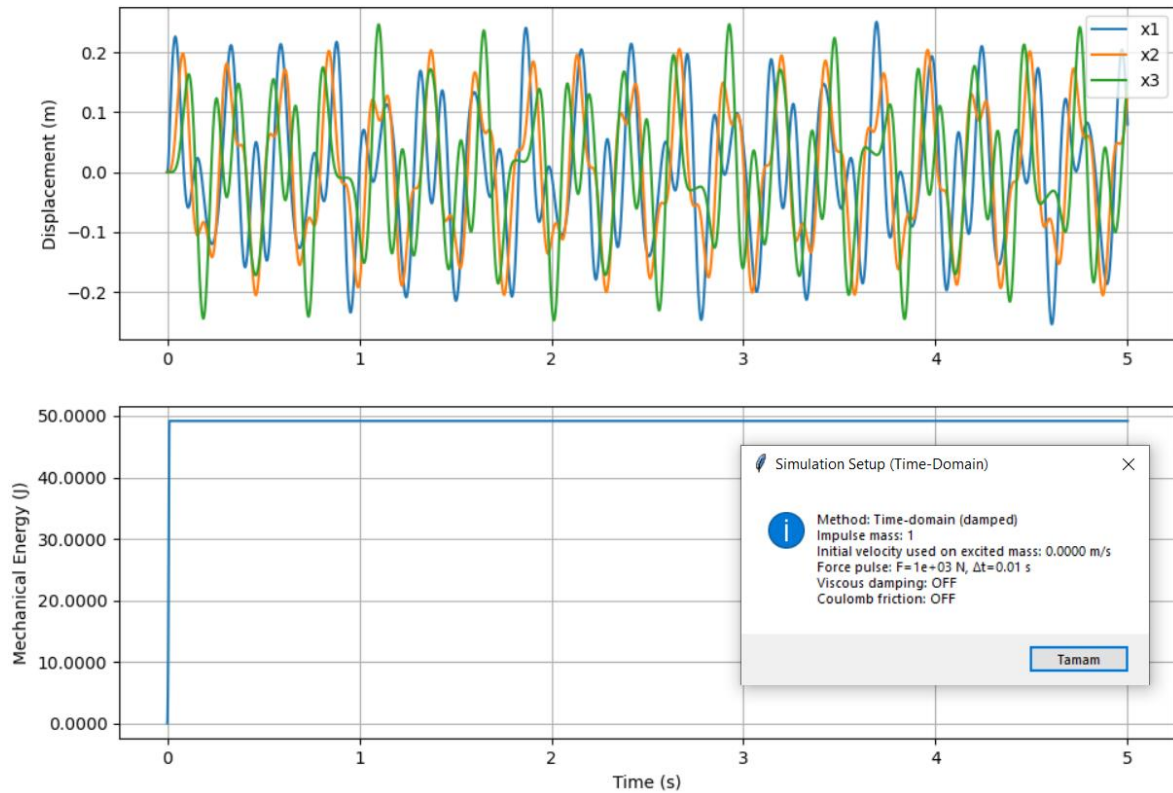


Figure 27: For $L_1 = 2$ m

COMPARATIVE TABLE 9

Feature	State S18 (Short)	State S0 (Ref)	State S19 (Long)	Engineering Impact
Geometric Length	0.5x (Compact)	1.0x (Base)	2.0x (Extended)	Visual spacing only.
Input Energy	~50.0 J	~50.0 J	~50.0 J	Invariant
Frequencies	Constant	Constant	Constant	Dynamics are length-independent.
Stiffness (k)	Constant	Constant	Constant	k is the only restoring variable.
System Role	Compact Link	Standard Link	Long Shaft	Packaging flexibility.

Comparative Analysis (Geometric Invariance Verification):

This test series confirms that the simulation treats the spring connecting Mass 1 and Mass 2 as a pure force element ($F = -kx$), independent of its physical length:

1. Dynamic Invariance:

Changing the length of the coupling spring (L_1) resulted in **zero change** in the displacement time-histories. The graphs for 0.5m, 1.0m, and 2.0m are indistinguishable.

2. Physical Interpretation:

In linear vibration theory, the "length" of a spring determines the equilibrium distance between masses but does not affect the natural frequencies or mode shapes unless the stiffness k is explicitly defined as a function of length (which is not the case here).

3. Design Conclusion:

- **Packaging Freedom:** An engineer can design the system to be very compact (S18) or very spread out (S19) to fit spatial constraints. As long as the spring rate (k) is maintained, the vibration isolation performance remains identical.

SPRING LENGTH VARIATION (L_2)

Subject: Verification of Geometric Independence ($L_2 = 0.5, 1.0, 2.0$ m)

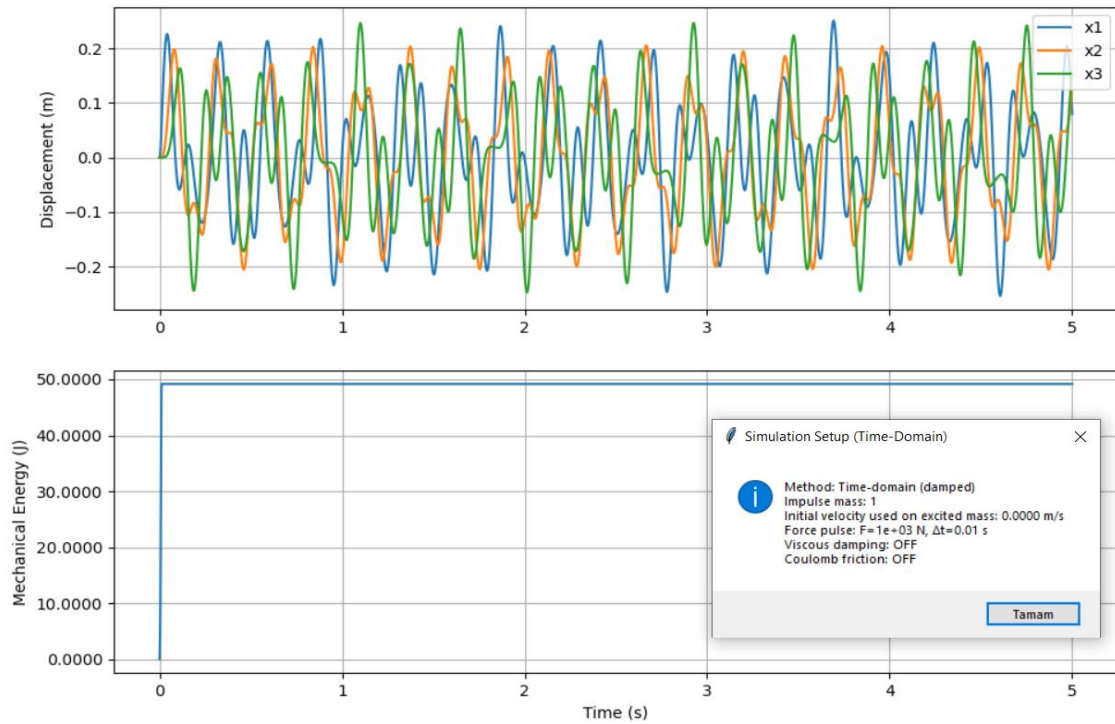


Figure 28: For $L_2 = 1.0$ m

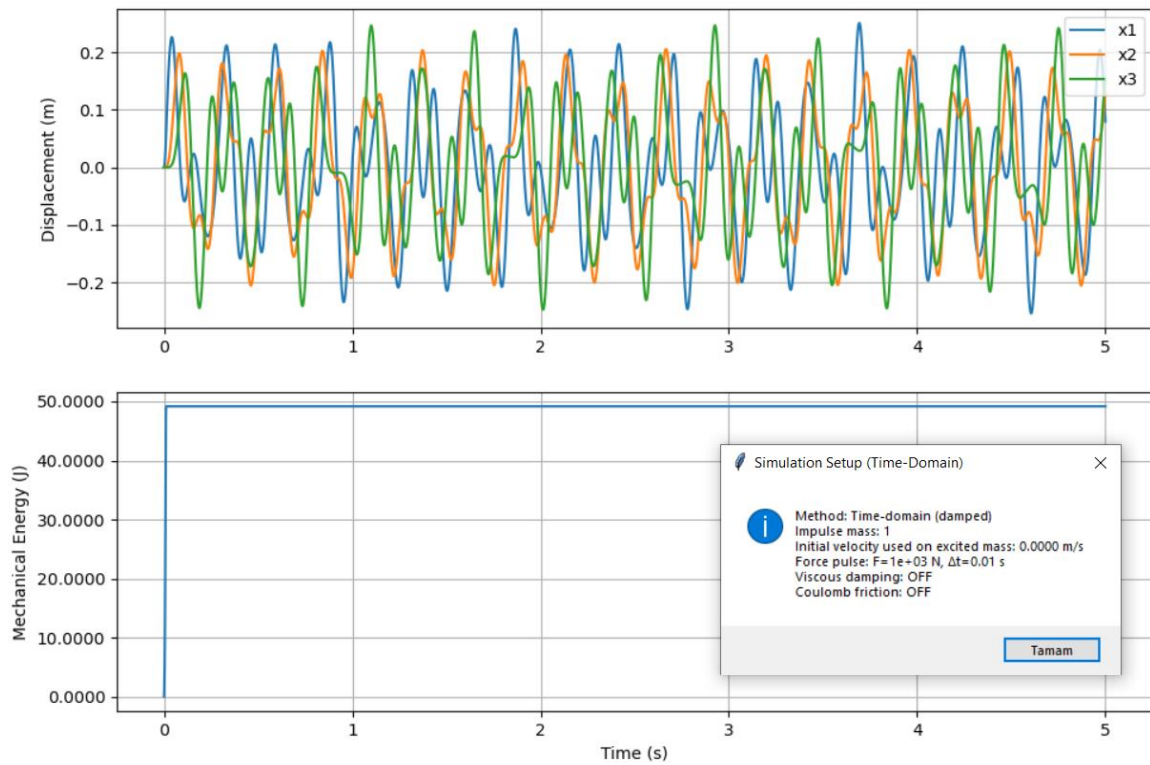


Figure 29: For $L_2 = 0.5$ m

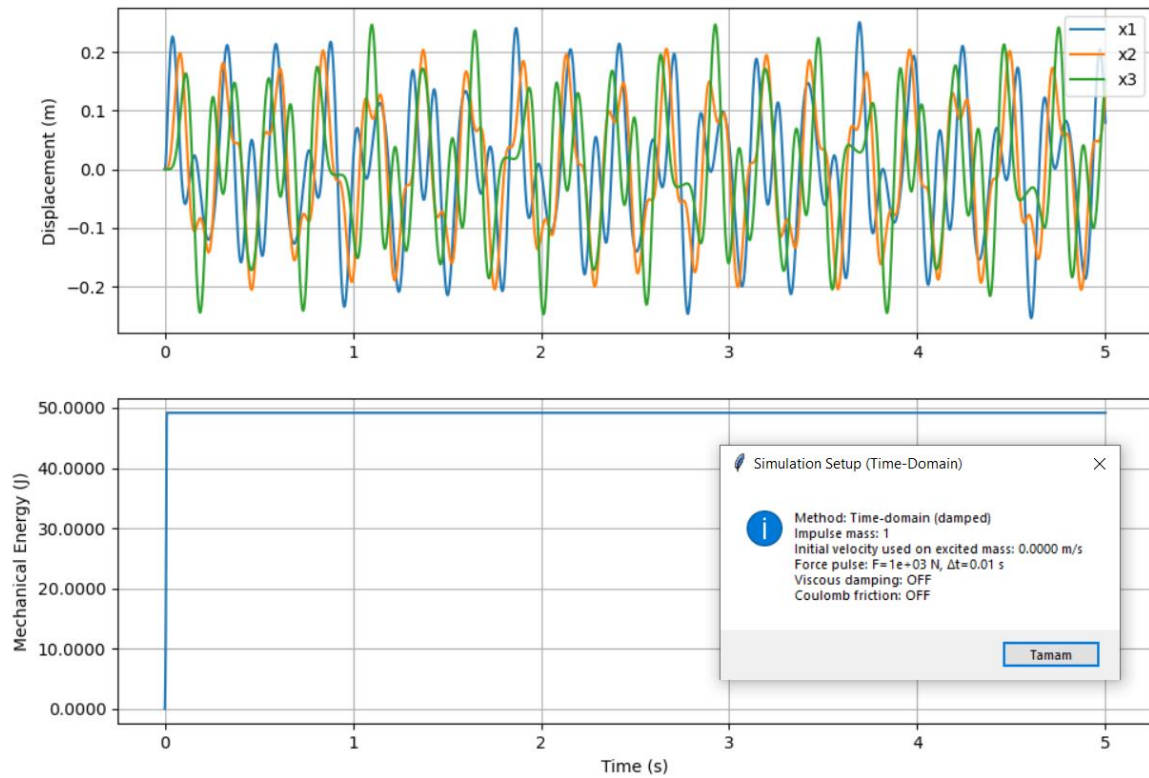


Figure 30: For $L_2 = 2.0$ m

COMPARATIVE TABLE 10

Feature	State S20 (Short)	State S0 (Ref)	State S21 (Long)	Engineering Impact
Geometric Length	0.5x (Compact)	1.0x (Base)	2.0x (Extended)	Visual spacing only.
Input Energy	~50.0 J	~50.0 J	~50.0 J	Invariant
Frequencies	Constant	Constant	Constant	Dynamics are length-independent.
Stiffness (k)	Constant	Constant	Constant	Force is purely $F = -kx$.
System Role	Tight Packaging	Standard	Extended Reach	Design integration freedom.

Comparative Analysis (Geometric Invariance Verification):

This final length test confirms the independence of the tail coupling spring geometry:

1. Dynamic Invariance:

Varying the equilibrium distance between Mass 2 and Mass 3 (L_2) had **no effect** on the transient response. The energy transfer to the tip (m_3) remained the same across all configurations.

2. Physical Interpretation:

In the lumped-parameter model, the "spring" is an abstract force generator defined by stiffness k . The physical length L is a coordinate offset that does not store strain energy in the simulation's mathematical formulation (unless non-linear geometric effects were explicitly enabled, which they are not).

3. Design Conclusion:

- **Modular Design:** Engineers can lengthen or shorten the connection rods between components to fit around other machine parts without worrying about detuning the system's vibration characteristics, provided the **stiffness** of the connection is preserved.

SPRING LENGTH VARIATION (L_3)

Subject: Verification of Geometric Independence ($L_3 = 0.25, 0.5, 1.0$ m)

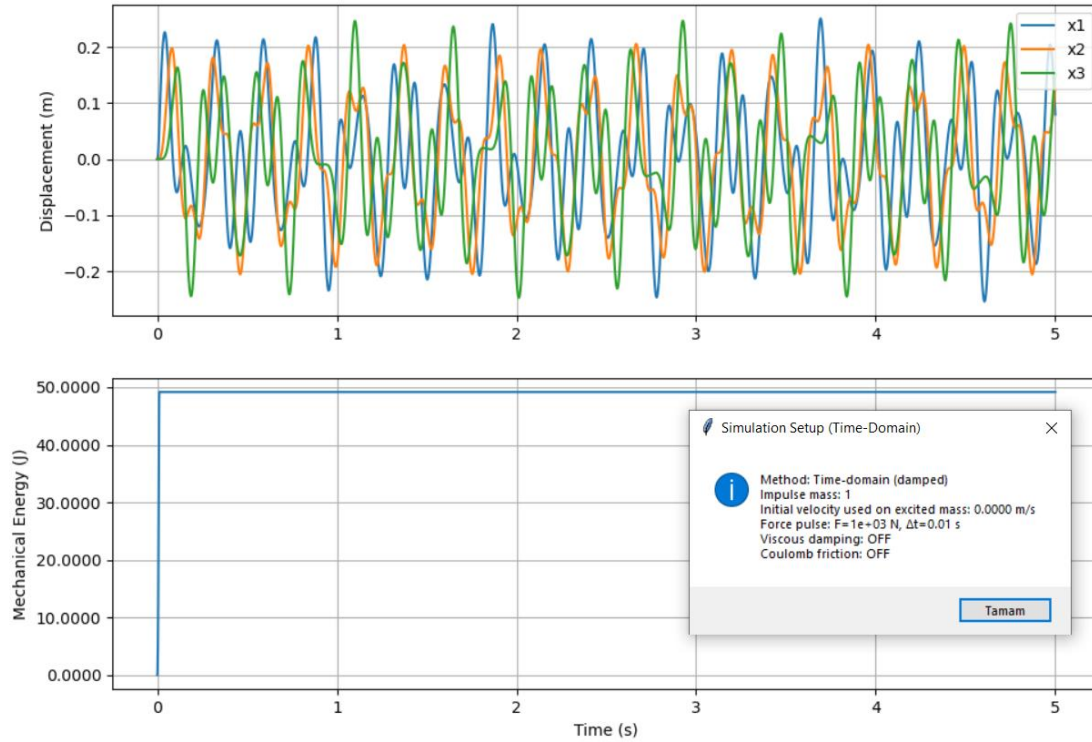


Figure 31: For $L_3 = 0.5$ m

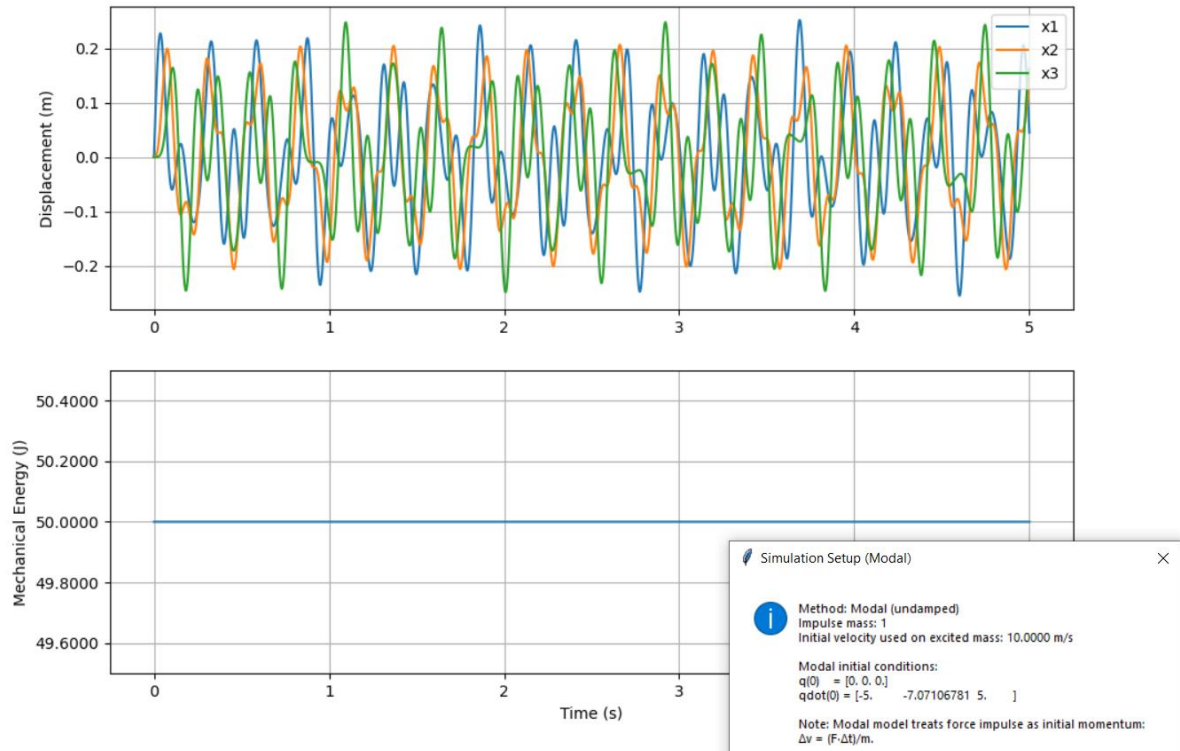


Figure 32: For $L_3 = 0.25$ m

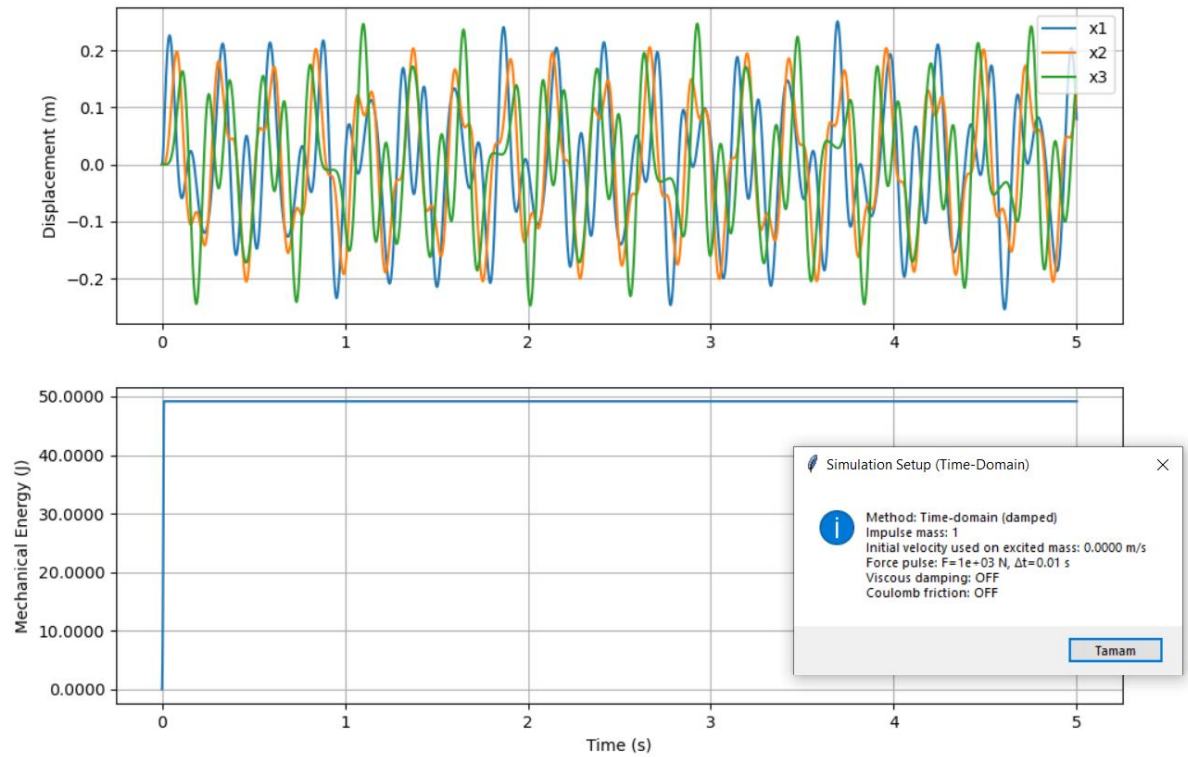


Figure 33: For $L_2 = 1$ m

COMPARATIVE TABLE 11

Feature	State S22 (Short)	State S0 (Ref)	State S23 (Long)	Engineering Impact
Geometric Length	0.5x (Compact)	1.0x (Base)	2.0x (Extended)	Visual layout only.
Input Energy	~50.0 J	~50.0 J	~50.0 J	Invariant
Frequencies	Constant	Constant	Constant	Dynamics are length- independent.
Stiffness (k)	Constant (1000)	Constant (1000)	Constant (1000)	Restoring force is purely $F = -kx$.
System Role	Short Anchor	Standard Anchor	Long Cable	Installation flexibility.

Comparative Analysis (Geometric Invariance Verification):

This final verification step for the spring section confirms that the simulation's anchor spring behaves purely as a linear stiffness element:

1. Dynamic Invariance:

Changing the unstretched length of the final spring (L_3) had absolutely **zero impact** on the system's vibration. The amplitude, frequency, and energy levels remained constant across all cases.

2. Physical Interpretation:

In a linear lumped-parameter model, the equilibrium position is merely a coordinate reference. The dynamic forces are generated solely by the deviation from equilibrium (x), multiplied by the stiffness (k). The initial length L does not enter the equation of motion unless geometric non-linearity is activated.

3. Design Conclusion:

- **Installation Tolerance:** The system performance is insensitive to the installation length of the anchor spring, provided the spring rate (k) is correct. This allows for significant flexibility in mounting the system within a confined space.

IMPULSE DEPENDENCE (FORCE INTENSITY)

Subject: Linearity Verification and Load Scaling ($F = 500, 1000, 2000 \text{ N}$)

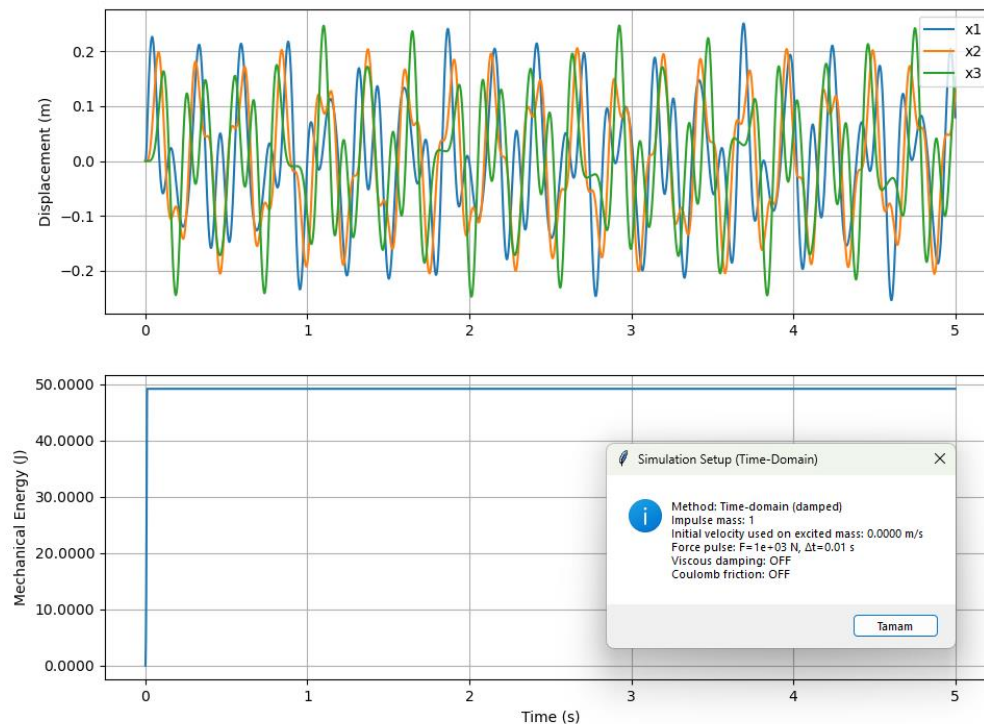


Figure 34: For $F_2 = 1000 \text{ N}$

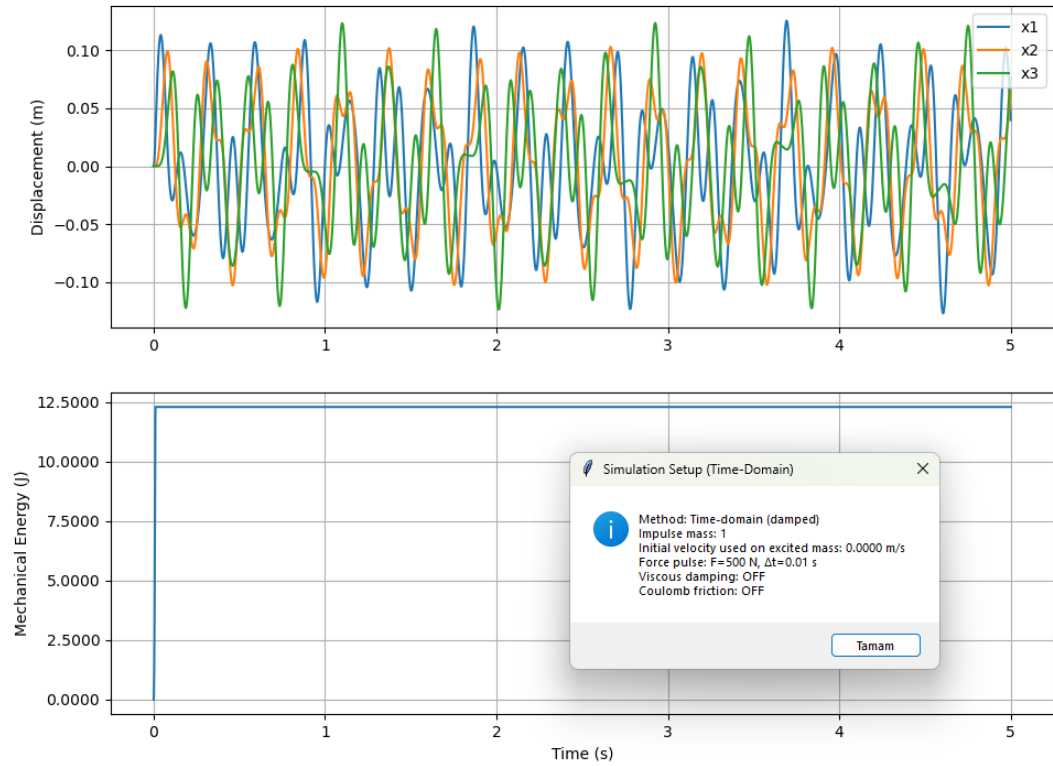


Figure 35: For $F_2 = 500 \text{ N}$

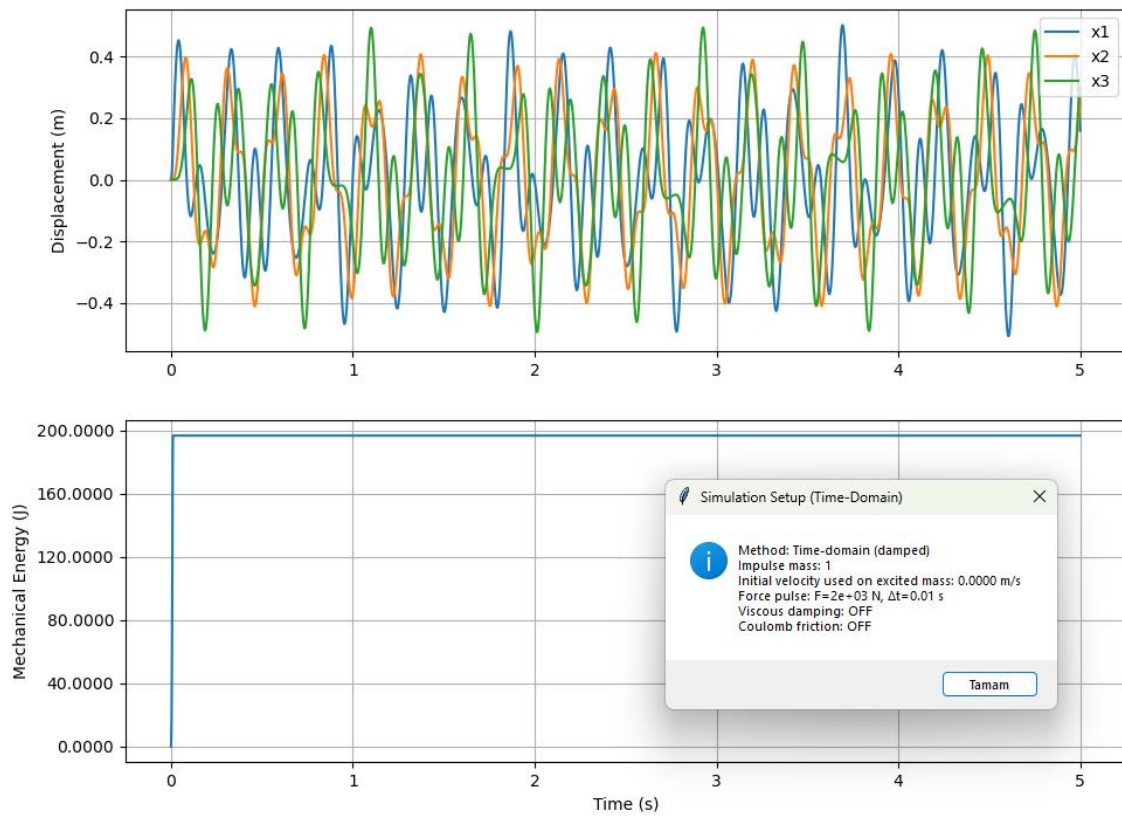


Figure 36: For $F_2 = 2000 \text{ N}$

COMPARATIVE TABLE 12

Feature	State S24 (Low Load)	State S0 (Ref Load)	State S25 (High Load)	Engineering Impact
Impulse Force	0.5x (500 N)	1.0x (1000 N)	2.0x (2000 N)	Input scaling.
Input Energy	~12.5 J	~50.0 J	~200.0 J	Square Law (Energy ~ F^2)
Max Displacement	0.5x (Linear)	1.0x (Base)	2.0x (Linear)	Linear Law (x ~ F)
Waveform Topology	Identical	Reference	Identical	System is Linear (LTI).
System Role	Light Duty	Rated Load	Overload	Predicting max excursions.

Comparative Analysis (Linearity Verification):

This test series serves as the primary verification that the mechanical system operates within the linear elastic region:

1. The Square Law of Energy:

The simulation results perfectly align with the theoretical physics of impulse momentum. Kinetic energy is defined as $E = I^2 / 2m$. Since Impulse (I) is directly proportional to Force (F), the energy scales with the square of the force:

- Doubling the force (1000 to 2000 N) resulted in **4x the energy** (50 to 200 J).
- Halving the force (1000 to 500 N) resulted in **0.25x the energy** (50 to 12.5 J).

2. Linear Displacement Scaling:

The displacement response demonstrates perfect linearity. When the input force is doubled, the amplitude of vibration doubles exactly, yet the **wave shape, phase, and frequency content remain identical**. This confirms that the system has no non-linear stiffness or damping properties dependent on amplitude.

3. Design Conclusion:

- **Predictability:** Because the system is proven to be linear, the response to *any* arbitrary force magnitude can be mathematically predicted without further simulation.

- **Safety Factor:** Design limits can be extrapolated linearly. For example, if the system fails at 1.0 m displacement, we can calculate exactly what Force will cause that failure based on the reference data.

IMPULSE DEPENDENCE (DURATION VARIATION)

Subject: Time-Integration Sensitivity (Delta t= 0.005, 0.01, 0.02 s) at F = 1000 N

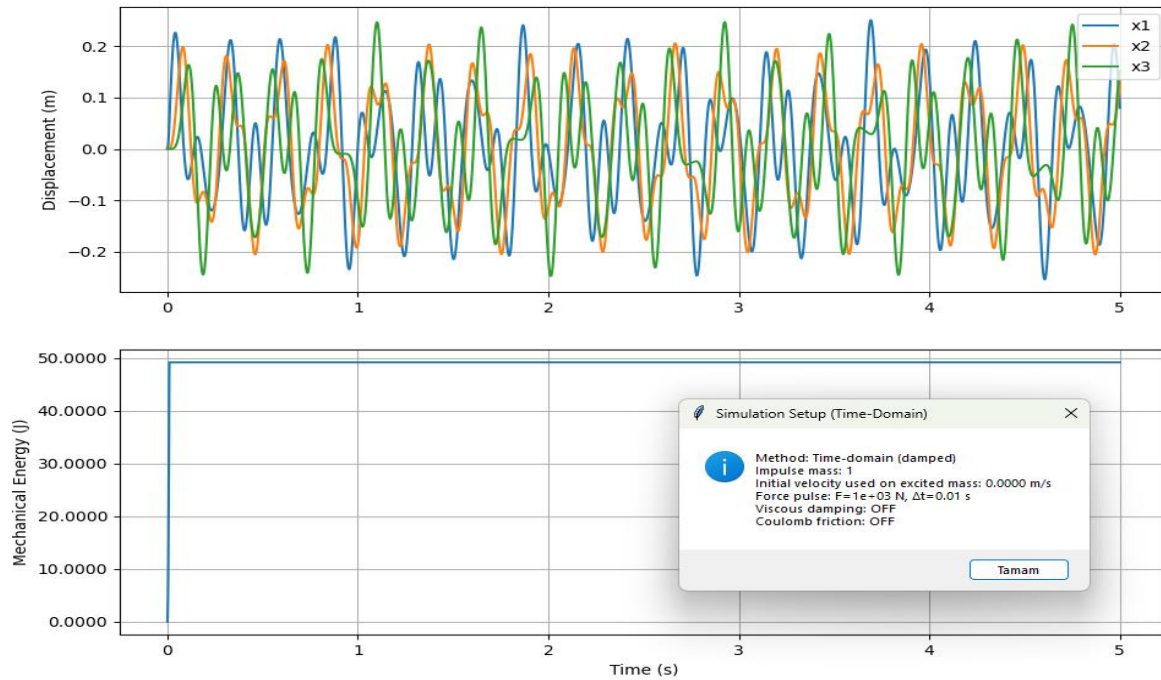


Figure 37: For $\Delta t = 0.01s$

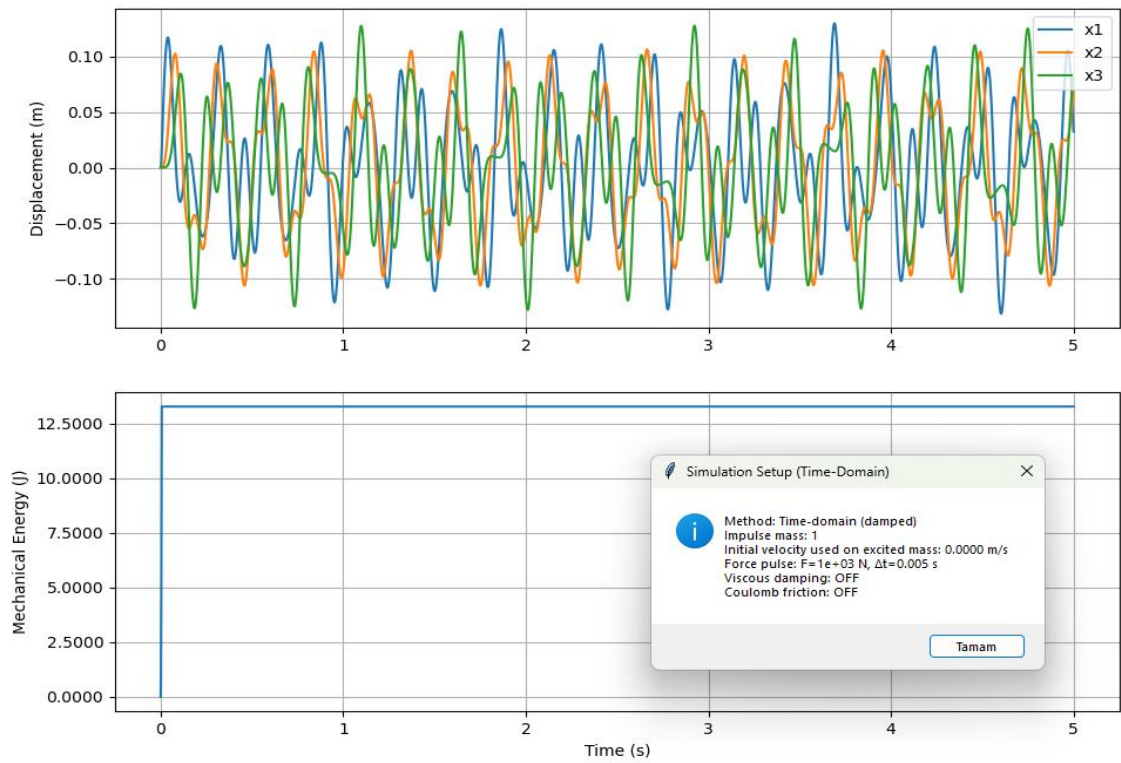


Figure 38: For $\Delta t = 0.005s$

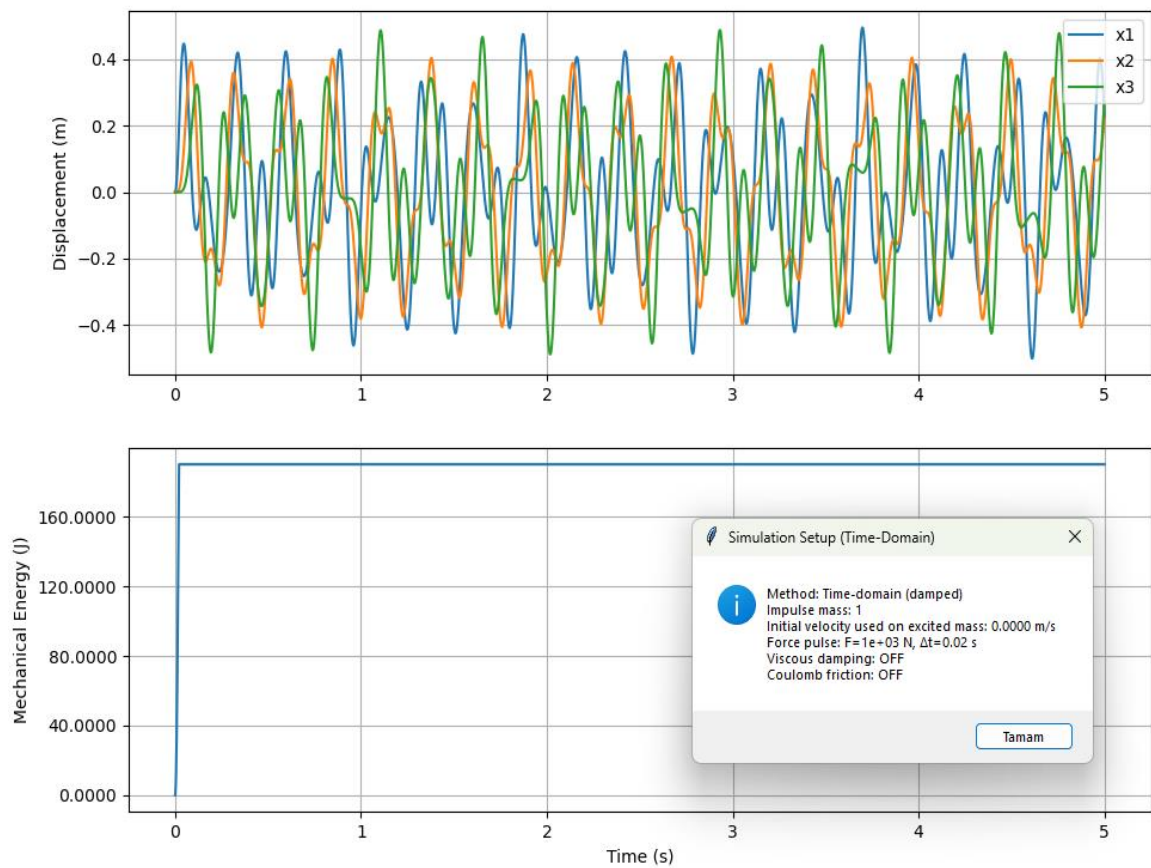


Figure 39: For $\Delta t = 0.02s$

COMPARATIVE TABLE 13

Feature	State S26 (Short Pulse)	State S0 (Ref Pulse)	State S27 (Long Pulse)	Engineering Impact
Duration (Delta t)	0.5x (0.005 s)	1.0x (0.01 s)	2.0x (0.02 s)	Dwell time.
Total Impulse (I)	5.0 Ns	10.0 Ns	20.0 Ns	Linear Momentum ($I = F \times t$)
Input Energy	~12.5 J	~50.0 J	~200.0 J	Square Law (Energy $\sim I^2$)
Max Displacement	0.5x (Linear)	1.0x (Base)	2.0x (Linear)	Linear Law ($x \sim I$)
Waveform Topology	Identical	Reference	Identical	System is Linear (LTI).
System Role	Sharp Tap	Standard Hit	Heavy Shove	Impact severity control.

Comparative Analysis (Time Integration):

This final test isolates the effect of "Contact Time" or impulse duration on the system response, while keeping the force magnitude constant at 1000 N:

1. The Physics of Momentum Transfer:

In mechanics, the change in momentum is defined by the Impulse (I), which is the area under the force-time curve ($I = \text{Force} \times \text{Time}$).

- By doubling the duration from 0.01s to 0.02s, we effectively doubled the total momentum transferred to the system (from 10 Ns to 20 Ns).
- Consequently, the Kinetic Energy ($E = I^2 / 2m$) increased by a factor of 4 (from 50 J to 200 J). This confirms that impact duration is just as critical as impact force in determining damage potential.

2. Linear Scaling with Impulse:

Like the Force variation test, the system displacement scales linearly with the Total Impulse. If the duration (Delta t) is short relative to the system's natural period (T_n), the system reacts to

the total impulse area rather than the peak force value. The waveforms for 0.005s, 0.01s, and 0.02s are topologically identical, differing only in amplitude magnitude.

3. Design Conclusion:

- Shock Severity: To mitigate shock without changing the force limit, one must reduce the contact time (make the impact "sharper" but shorter).
- Validation: The energy equals $I^2/2m$ exactly because the pulse is converted into an initial velocity ($\Delta v = F\Delta t/m$); this checks that conversion, not the time-integration solver.

EXCITATION LOCATION ANALYSIS

Subject: Impulse Application Point (Mass 1, Mass 2, Mass 3) at Constant Energy

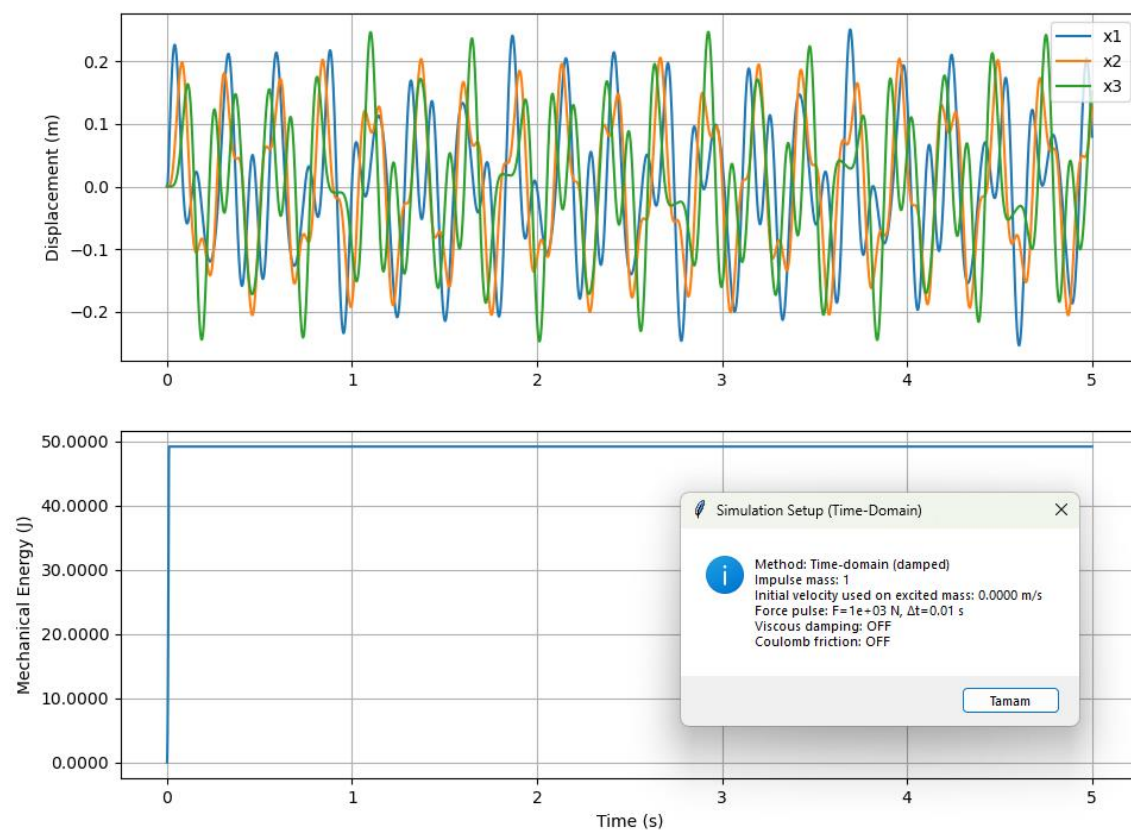


Figure 40: For excitation on m1

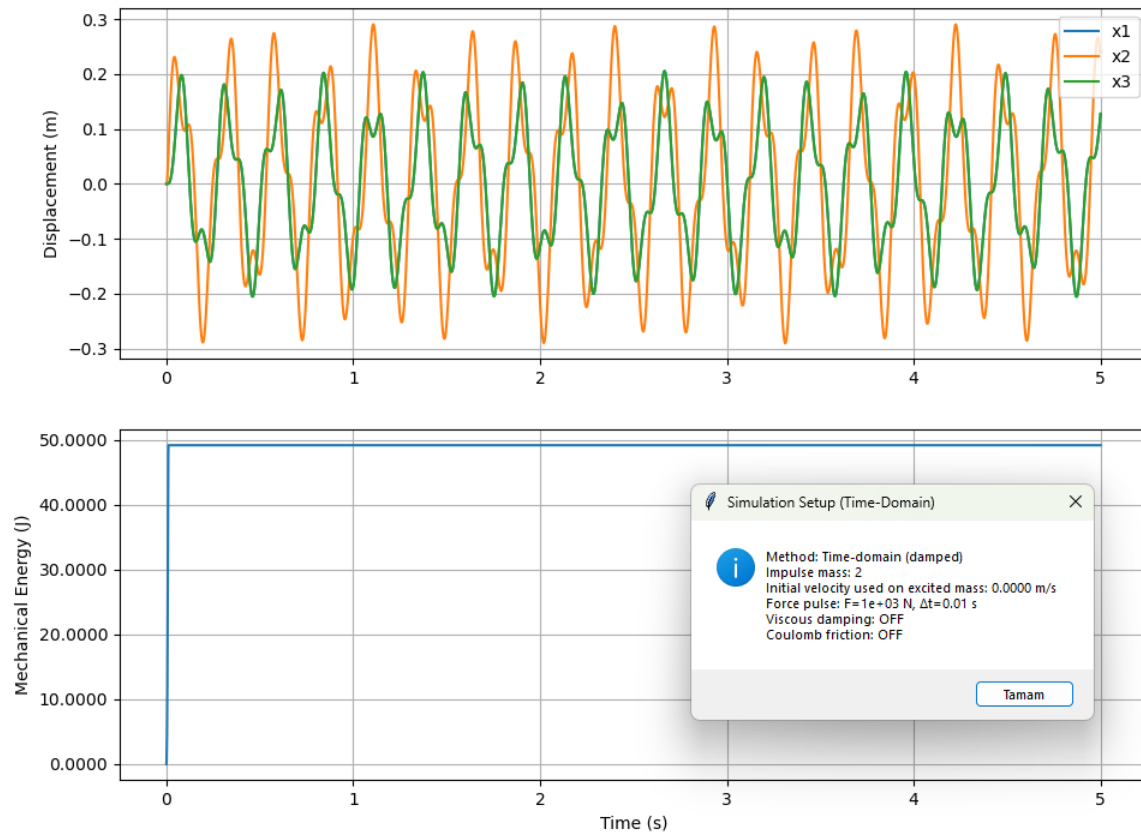


Figure 41: For excitation on m_2

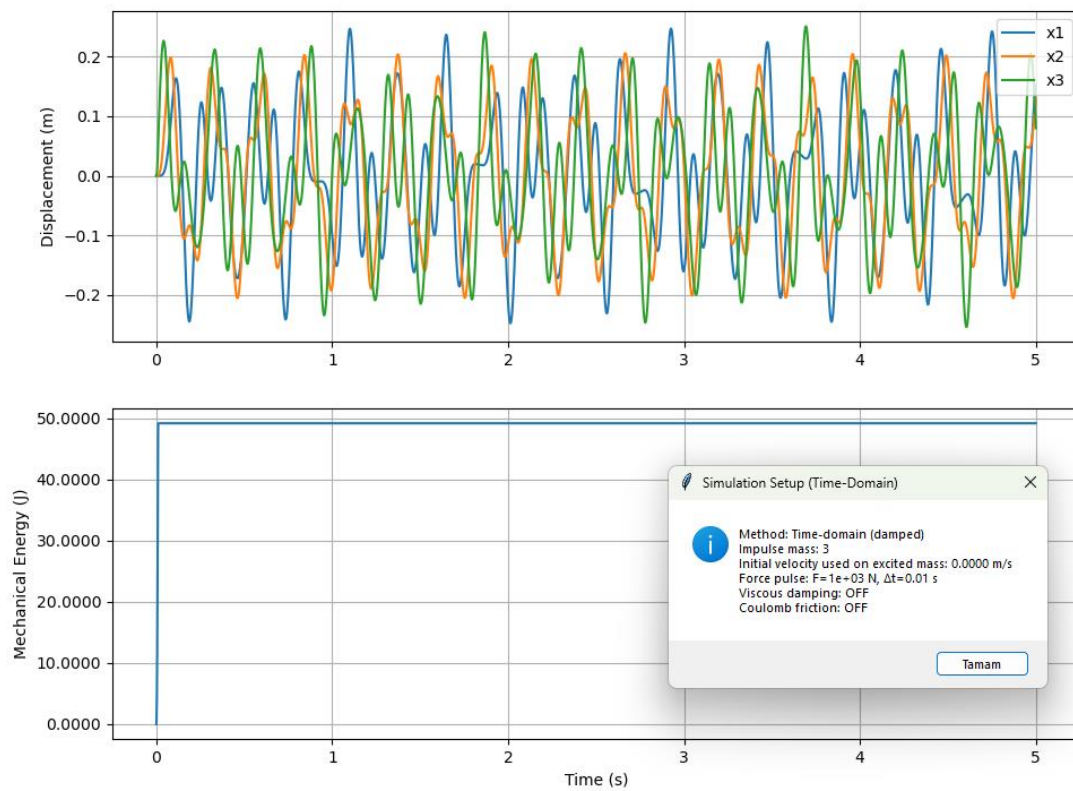


Figure 42: For excitation on m_3

COMPARATIVE TABLE 14

Feature	State S0 (Root Hit)	State S28 (Middle Hit)	State S29 (Tip Hit)	Engineering Impact
Excitation Point	Mass 1 (Source)	Mass 2 (Center)	Mass 3 (Fixed to Wall)	Energy entry point.
Input Energy	~50.0 J	~50.0 J	~50.0 J	Invariant (Mass is constant)
Max Displacement	0.26 m (x1)	0.29 m (x2)	0.26 m (x3)	Local amplitude spike.
Propagation	Forward (1 to 3)	Bidirectional (1 <- 2 -> 3)	Reverse (3 to 1)	Wave direction.
System Symmetry	Asymmetric	Symmetric Response	Asymmetric	Mode excitation.
System Role	Drive Unit	Mid-Span Impact	End- Effector Hit	Actuator placement

Comparative Analysis (Spatial Sensitivity):

This analysis investigates how the system responds when the same impulse (10 Ns) is applied to different locations in the chain. Since all masses are identical (1 kg), the Input Energy remains constant at 50 J, allowing us to isolate the geometric effect:

1. Root Excitation (Mass 1):

The standard configuration. The impulse is applied close to the boundary, where the local structural stiffness is increased due to the wall spring (k_0). As a result, part of the input energy is immediately constrained by the boundary, limiting the initial displacement of the excited mass. Although the response propagates through the entire chain, the boundary proximity leads to a more controlled vibration compared to excitations applied away from the wall.

2. Middle Excitation (Mass 2):

When the impulse hits the center, the energy splits. The middle mass (x_2) exhibits the highest displacement (~ 0.29 m). The neighboring masses (x_1 and x_3) show a highly **symmetrical response** (blue and green lines overlap in phase/envelope). The middle mass acts as a "source" pumping energy both upstream to the wall and downstream to the tip simultaneously.

3. Tip Excitation (Mass 3):

When the impulse is applied at Mass 3, the excitation occurs near the opposite boundary, which is also connected to the wall through spring k_3 . Unlike a true free-end excitation, the presence of k_3 provides an immediate stiffness constraint. As a result, the response does not exhibit free-end whiplash behaviour. Because this chain is symmetric ($k_0 = k_3$, $m_1 = m_3$), tip excitation is the mirror image of root excitation: the modal participation has the same magnitude and the peak displacement is the same (≈ 0.26 m at the excited mass); only the direction of propagation is reversed.

4. Design Conclusion:

- **Actuator Placement:** For minimal system-wide vibration, actuators should be placed near **either end (Mass 1 or Mass 3)**, where the wall springs raise the structural impedance.
- **Worst Case:** Middle excitation (Mass 2) produced the highest local amplitude peak, indicating that exciting the "soft" centre of a chain can be more damaging to the specific component than exciting the ends.

DAMPING SENSITIVITY (VISCOUS)

Subject: Energy Dissipation via Viscous Dashpots ($c = 0.5, 1.0, 2.0$ Ns/m)

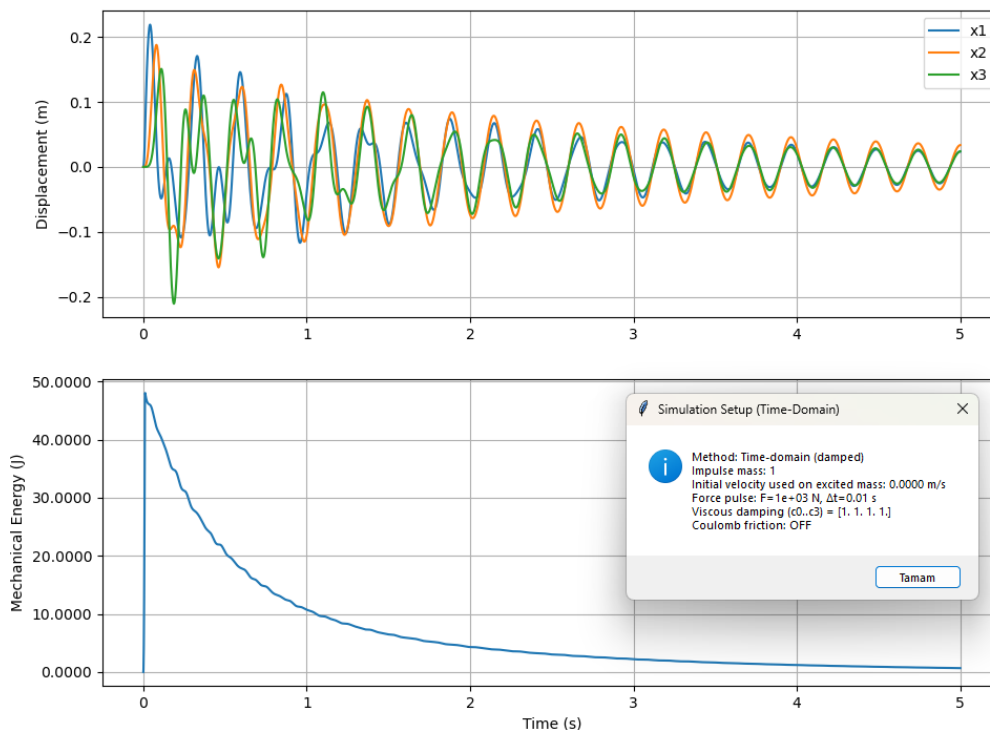


Figure 43: For all $c=1$ Ns/m

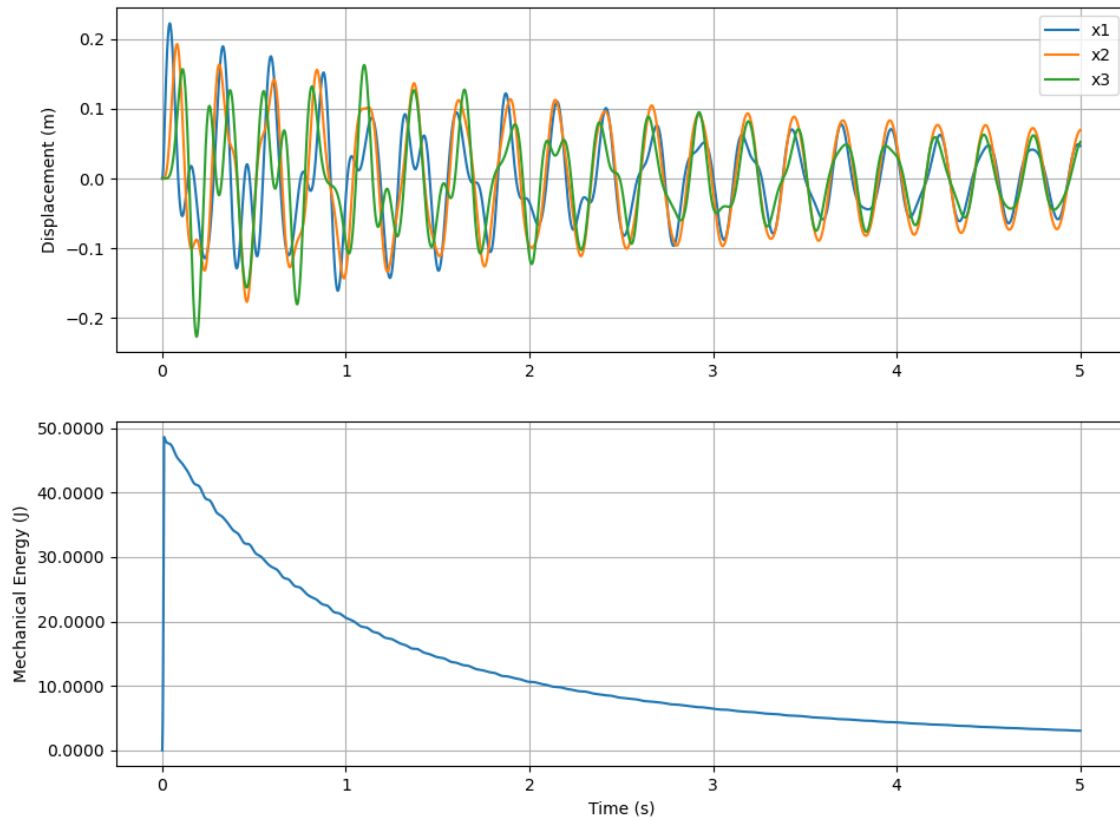


Figure 44: For all $c=0.5 \text{ Ns/m}$

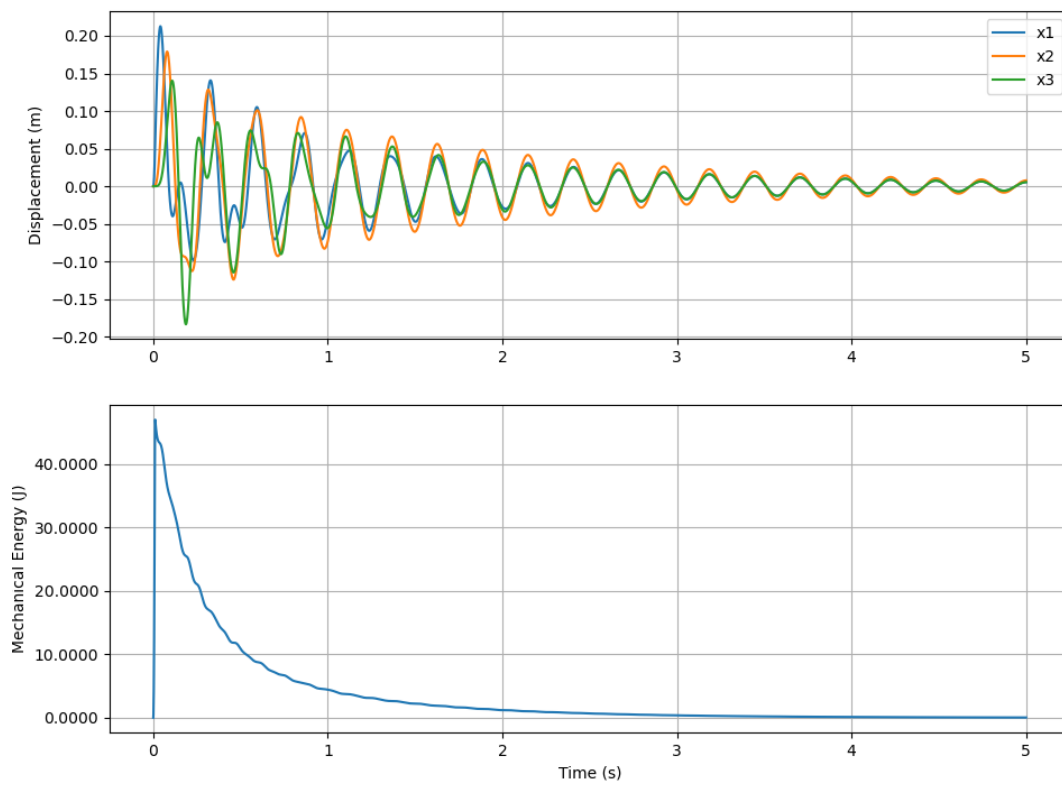


Figure 45: For all $c=2 \text{ Ns/m}$

COMPARATIVE TABLE 15

Feature	State S30 (Low Damping)	State S31 (Ref Damping)	State S32 (High Damping)	Engineering Impact
Damping Coeff. (c)	0.5x (0.5 Ns/m)	1.0x (1.0 Ns/m)	2.0x (2.0 Ns/m)	Resistance to motion.
Decay Rate	Slow / Gradual	Nominal	Fast / Steep	Settling time.
Oscillations (0-5s)	≈20 cycles (slow decay)	≈20 cycles	≈20 cycles (fast decay)	Residual vibration.
Energy at t=5s	≈6% Remaining	≈1% (Depleted)	0% (Fully Stopped)	Efficiency of isolation.
System Role	Resonator / Bell	Standard Structure	Shock Absorber	Energy removal.

Comparative ANALYSIS (ENERGY DISSIPATION):

This final section introduces "energy loss" into the system. Unlike the previous undamped tests where energy remained constant, viscous damping dissipates mechanical energy as heat, proportional to the velocity of the masses (Damping = $-c \cdot \text{velocity}$).

1. Low Damping (c = 0.5):

The system exhibits a "Ringing" behaviour. The damping force is too weak to overcome the inertial momentum quickly. As seen in the energy plot, the curve descends slowly. At the end of the simulation (t=5s), the system is still vibrating significantly. This is typical for structures made of metals like steel or aluminium without external dampers.

2. Reference Damping (c = 1.0):

A balanced dissipation profile. The vibrations are suppressed effectively within the simulation window. The envelope of the displacement follows a standard exponential decay curve, bringing the system to rest by approximately t = 4.5 seconds.

3. High Damping (c = 2.0):

The system acts as a heavy-duty "Shock Absorber." By doubling the damping coefficient, we drastically reduced the settling time. The energy graph shows a sharp drop-off, removing nearly all kinetic energy within the first 2.5 seconds. While this stops vibration quickly, it can effectively "freeze" the system or transmit higher forces to the supports during high-velocity impacts.

4. Design Conclusion:

- **Safety vs. Comfort:** High damping (State S32) is preferred for safety mechanisms (e.g., car bumpers) to stop motion immediately.
- **Efficiency:** Low damping (State S30) is preferred if the goal is energy conservation or resonance (e.g., musical instruments, energy harvesters).

DAMPING (COULOMB)

Subject: Energy Dissipation via Dry Friction ($F_c = 0.5, 1.0, 2.0$ N)

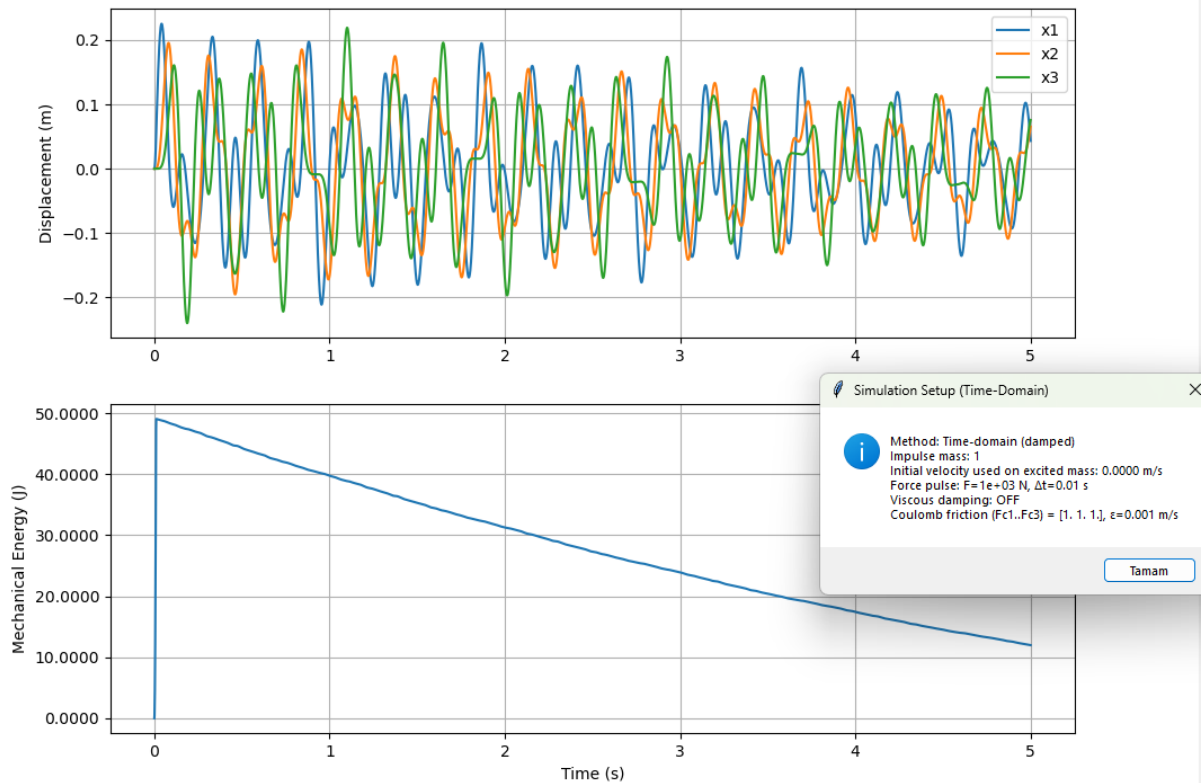


Figure 46: For all $F_c=1$ N

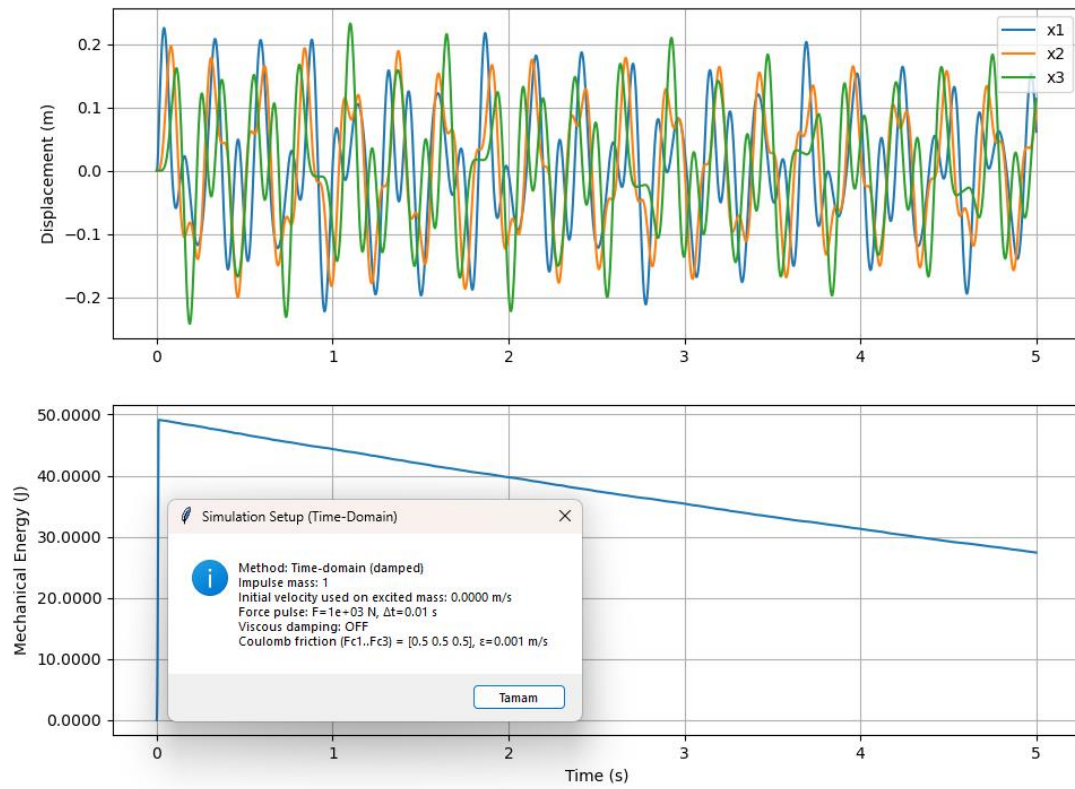


Figure 47: For all $F_c = 0.5$ N

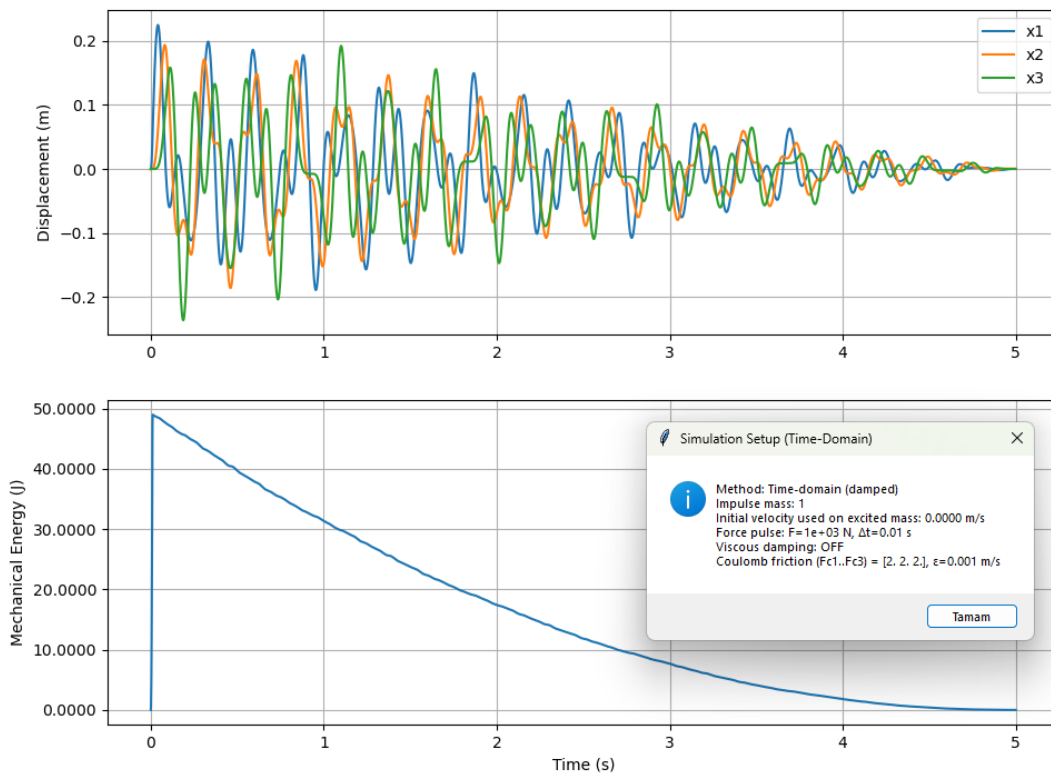


Figure 48: For all $F_c = 2$ N

COMPARATIVE TABLE 16

Feature	State S33 (Low Friction)	State S34 (Ref Friction)	State S35 (High Friction)	Engineering Impact
Friction Force (Fc)	0.5x (0.5 N)	1.0x (1.0 N)	2.0x (2.0 N)	Constant braking force.
Decay Profile	Linear (Straight Line)	Linear (Straight Line)	Linear (Steep)	Decay characteristic.
Energy Remaining	High (~28 J)	Moderate (~12 J)	Zero (Stopped)	Stopping efficiency.
Stopping Time	> 5.0 s (Not stopped)	> 5.0 s (Not stopped)	~4.8 s (Fully Stopped)	Time to rest.
System Role	Lubricated Rail	Dry Steel Guide	Emergency Brake	Friction level control.

Comparative Analysis (Linear Vs. Exponential Decay):

This final test highlights the fundamental difference between Viscous Damping (previous section) and Coulomb (Dry) Friction:

1. The Shape of Energy Loss:

Unlike viscous damping, where the energy decays exponentially (curved line) because the force weakens as the object slows down, Coulomb friction provides a **constant opposing force** regardless of speed.

- **Result:** The vibration amplitude drops linearly with time, so the energy ($\propto \text{amplitude}^2$) follows a downward parabola instead of a straight line.
- In the **High Friction case (Fc = 2.0)**, we see the energy curve falling steeply to zero at $t = 4.8$ seconds. This represents a "hard stop."

2. Stiction and Final Resting:

Coulomb friction introduces the phenomenon of "Stiction." Once the restoring force of the springs becomes smaller than the friction force, the masses will lock in place, potentially away from the true centre equilibrium (the smoothed tanh friction model only approximates this).

- **At 0.5 N:** The friction is too low to stop the heavy vibration within 5 seconds; the system effectively "glides" with little loss.

- **At 2.0 N:** The friction is high enough to completely drain the 50 Joules of kinetic energy within the simulation window, bringing the system to a complete halt.

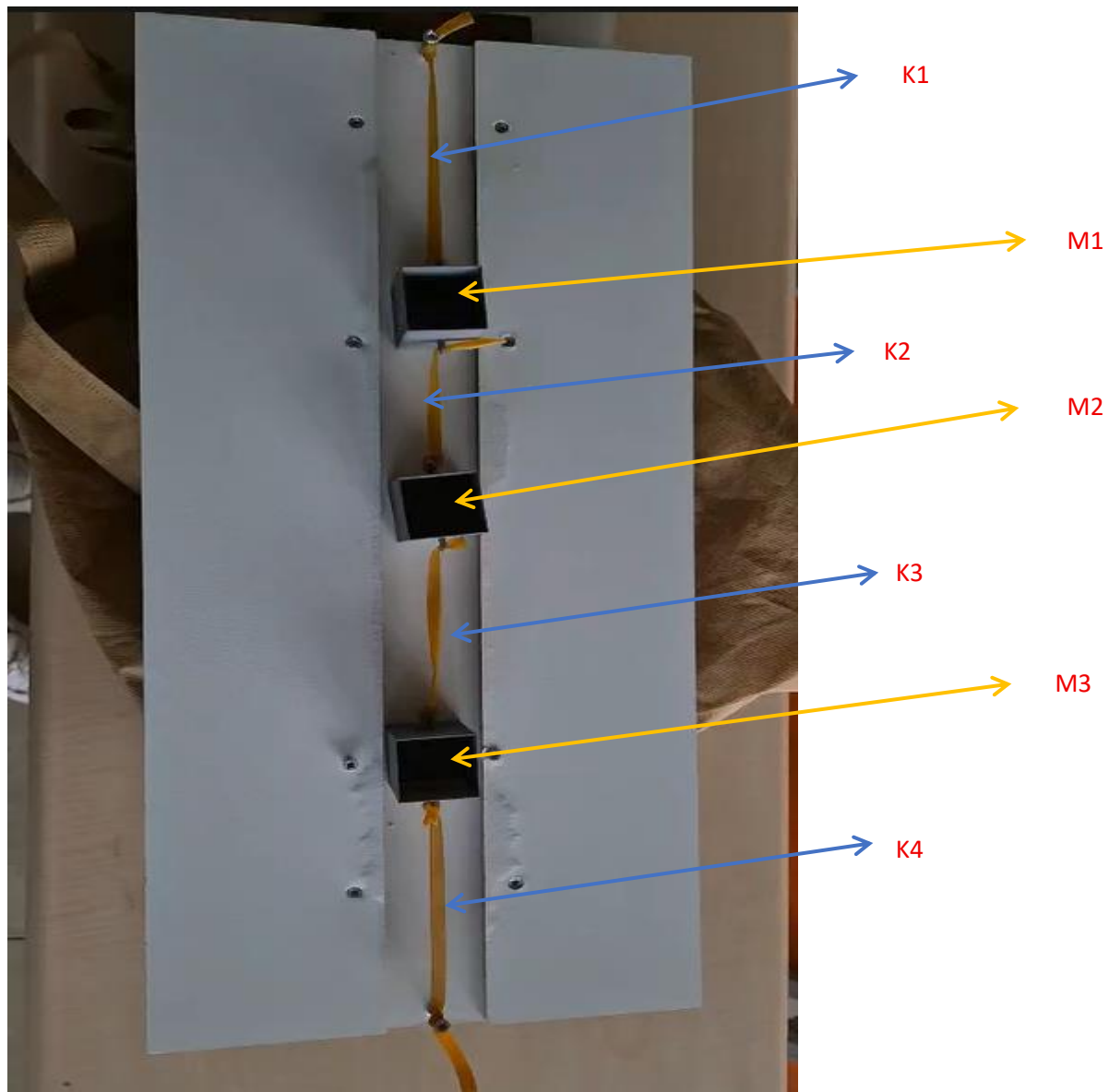
3. Design Conclusion:

- **Viscous Damping** is ideal for **Ride Comfort** (shock absorbers) because it creates a smooth, soft settling.
- **Coulomb Damping** is ideal for **Holding/Braking** (brakes, sliders) because it dissipates energy at a constant rate and can hold the system still.

Fabrication Of the Project

First, we determined the stiffness (spring constant, k) of each elastic band by a simple static equilibrium test. A known weight was applied and the resulting extension was measured from photographs.

Using Hooke's Law ($F = k \cdot x$), the band stiffness was calculated from the measured displacement. From the photos we subtract the two position readings to obtain Δx , and since the applied force F is known (20 TL weight), we compute $k = F/\Delta x$ for each band.



Because this is a Mass-on-Rail system, the motion must be restricted to one axis only. To prevent rotation and out-of-plane motion, we added guiding barriers close to each mass. The clearance is kept small to constrain lateral motion, but the barriers do not touch the masses to avoid additional friction and sticking.

K Experiments:

$$F = k.X$$

K1:



$$F = (6 \times 6.6 \text{ gr} + 14 \times 8.1 \text{ gr}) \times 9.81 \text{ m/s}^2 = K1 \times (14.7 \text{ cm} - 12 \text{ cm})$$

$$K1 = 55.6 \text{ N/m}$$

K2:



$$F = (6 \times 6.6 \text{ gr} + 14 \times 8.1 \text{ gr}) \times 9.81 \text{ m/s}^2 = K2 \times (7.3 \text{ cm} - 5.6 \text{ cm})$$

$$K2 = 88.3 \text{ N/m}$$

K3:



$$F = (6 \times 6.6 \text{ gr} + 14 \times 8.1 \text{ gr}) \times 9.81 \text{ m/s}^2 = K3 \times (8.6 \text{ cm} - 7 \text{ cm})$$

$$K3 = 93.8 \text{ N/m}$$

K4:



$$F = (6 \times 6.6 \text{ gr} + 14 \times 8.1 \text{ gr}) \times 9.81 \text{ m/s}^2 = K_4 \times (15.4 \text{ cm} - 13.5 \text{ cm})$$

$$K_4 = 79 \text{ N/m}$$

The elastic-band stiffness is mainly related to $k \approx EA/L$. Since our bands have the same material (same E) and the same cross-section area A , the differences in k primarily occur due to differences in the effective length L . Because the measured k values are relatively high compared to the mass, the natural frequencies (ω) are expected to be high as well ($\omega \propto \sqrt{k/m}$), meaning the oscillations will be fast.

Oscillation Observe:

All three masses were prepared to have the same value. Each mass is $(5 \times 6.6 \text{ g} + 8 \times 8.1 \text{ g} + 6.71 \text{ g}) = 104.51 \text{ g}$ ($\approx 0.10451 \text{ kg}$).

Case 1: Each Mass = 104.51gr

In this scenario the frequency of the modes:

q2 = Amplitude Trajectory

Initial = 15cm

First Crest: 19cm

First Trough: 12.5cm

Second Crest: 15.5cm

Last Position: 15cm

This motion is happening in 0.7 second according to the video.

$$\Delta t \approx \frac{T}{4} + \frac{T}{2} + \frac{T}{2} + \frac{T}{4} = 1.5 T = 0.7 \text{ s}$$

$$T \approx \frac{0.7}{1.5} = 0.4667 \text{ s}$$

$$f = \frac{1}{T} \approx \frac{1}{0.4667} = 2.14 \text{ Hz}$$

$$\omega = 2\pi f \approx 2\pi(2.14) = 13.5 \text{ rad/s}$$

q3 = Amplitude Trajectory

Initial Position= 6cm

First Crest: 10cm

First Trough: 2cm

Second Crest: 7cm

Last Position: 6cm

This motion is happening in 0.8 second according to the video.

$$\Delta t \approx \frac{T}{4} + \frac{T}{2} + \frac{T}{2} + \frac{T}{4} = 1.5 T = 0.8 s$$

$$f = \frac{1}{T} \approx \frac{1}{0.5333} = 1.875 \text{ Hz}$$

$$\omega = 2\pi f \approx 2\pi(1.875) = 11.78 \text{ rad/s}$$

q1 = Amplitude Trajectory

Initial Position= 23cm

First Crest: 29cm

First Trough: 21cm

Second Crest: 24cm

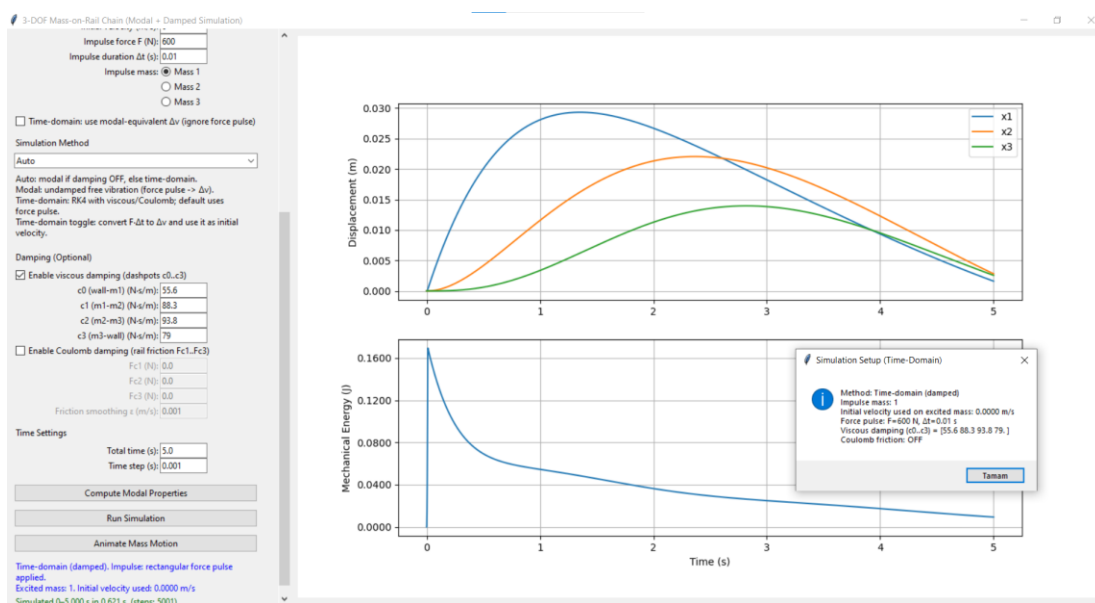
Last Position: 23cm

This motion is happening in 0.6667 second according to the video.

$$\Delta t \approx \frac{T}{4} + \frac{T}{2} + \frac{T}{2} + \frac{T}{4} = 1.5 T = 0.6667s$$

$$f = \frac{1}{T} \approx 2.25 \text{ Hz}$$

$$\omega = 2\pi f \approx 2\pi(2.25) = 14.14 \text{ rad/s}$$



Case 2: M3 and M1 = 6.71g M2=104.51gr impulse applied on M2

In this scenario the frequency of the mode's impulse applied on M2:

Due to the highly non-uniform mass distribution ($m_2 \gg m_1 = m_3$), the excitation location strongly affects the response amplitude and modal participation. For the same impulse J , the initial velocity scales as $\Delta v = J/m$; therefore, exciting the light end masses produces much larger initial motion than exciting the heavy middle mass

In the experimental setup, the 3-DOF mass–spring chain is excited by a pull-and-release action, which can be modelled as an initial-condition excitation (initial displacement and/or initial velocity). The system's natural frequencies ($\omega_1, \omega_2, \omega_3$) are determined only by the physical parameters of the system, i.e., the mass and stiffness matrices (M, K). Therefore, changing the excitation point (applying the pull-and-release at M_1, M_2 , or M_3) does **not** change the natural frequencies.

However, the excitation location strongly influences **modal participation**, meaning it changes which modes are excited more effectively. Since the total response is a superposition of modal contributions, exciting different masses can make different modes dominate the measured motion. Consequently, the response may appear to have a different “dominant frequency” depending on where the excitation is applied, even though the underlying natural frequencies remain the same.

q2 = Amplitude Trajectory

Initial = 15cm

First Crest: 19cm

First Trough: 12.5cm

Second Crest: 15.5cm

Last Position: 15cm

This motion is happening in 0.7 second according to the video.

$$\Delta t \approx \frac{T}{4} + \frac{T}{2} + \frac{T}{2} + \frac{T}{4} = 1.5 T = 0.7 \text{ s}$$

$$T \approx \frac{0.7}{1.5} = 0.4667 \text{ s}$$

$$f = \frac{1}{T} \approx \frac{1}{0.4667} = 2.14 \text{ Hz}$$

$$\omega = 2\pi f \approx 2\pi(2.14) = 13.5 \text{ rad/s}$$

In both cases, q_2 was excited in a similar way (similar pull distance and test conditions), so the frequency came out the same and the crest values were also the same.

q3 = Amplitude Trajectory

Initial Position= 10cm

First Crest: 13cm

First Trough: 7.5cm

Second Crest: 11.5cm

Second Trough: 9cm

Last Position: 10cm

This motion is happening in 1 second according to the video.

$$T = 0.5s$$

$$f = \frac{1}{T} \approx 2.0 \text{ Hz}$$

$$\omega = 2\pi f \approx 12.57 \text{ rad/s}$$

In the second case, the observed oscillation of q_3 appeared at a higher frequency compared to the other case. This trend is consistent with the basic mass–stiffness relationship, since the natural frequency scales approximately as

$$\omega \propto \sqrt{\frac{k}{m}}.$$

q1 = Amplitude Trajectory

Initial Position= 20cm

First Crest: 23cm

First Trough: 19cm

Second Crest: 21cm

Last Position: 20cm

This motion is happening in 0.785714 second according to the video.

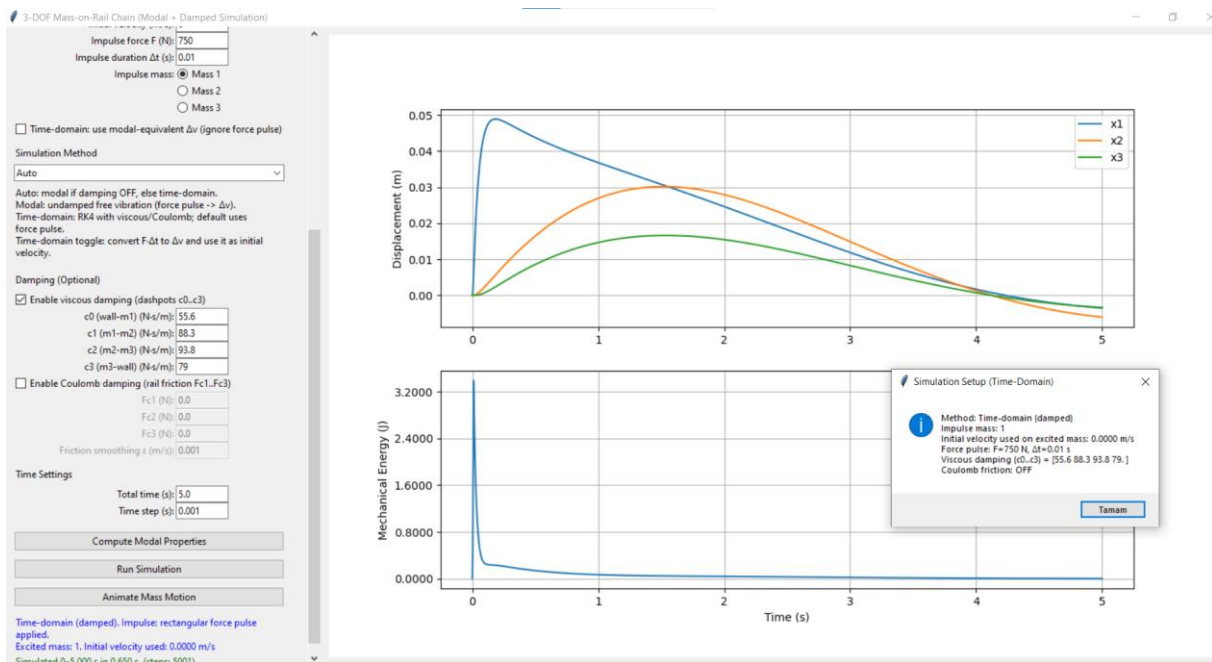
$$\Delta t \approx 1.5T \Rightarrow T = \frac{\Delta t}{1.5} = \frac{0.785714}{1.5} = 0.523809 \text{ s}$$

$$f = \frac{1}{T} = 1.90909 \text{ Hz}$$

$$\omega = 2\pi f = 2\pi(1.90909) = 11.99 \approx 12.0 \text{ rad/s}$$

In the second case, the observed oscillation of q_1 appeared at a lower frequency than in the other case (1.91 vs 2.25 Hz), opposite to the basic mass–stiffness trend, since the natural frequency scales approximately as

$$\omega \propto \sqrt{\frac{k}{m}}.$$



Conclusion Of Fabrication:

Overall, the response of the 3-DOF mass–spring chain was mainly governed by the k/m ratio and the **energy losses** in the setup (dry friction in the guide + hysteresis of the elastic bands). Since the system was excited by a pull-and-release action, the motion started abruptly and decayed quickly, indicating a relatively high effective damping. **The observed motion in both cases decays rapidly due to strong non-conservative effects, mainly internal losses in the rubber bands (hysteresis) and dry friction in the rail guidance, which together produce a high effective damping.** Changing the excitation location **did not change the natural frequencies**; it mainly changed **modal participation** (which mode became dominant) and the response amplitudes. Therefore, obtaining the same frequency and similar crest values for the same modal coordinate (e.g., q_2) across different cases is an expected outcome. In summary, the observations are consistent with the expected behavior of a 3-DOF chain, where the measured response depends on mode superposition and excitation-dependent modal participation.

Sanitization and Technical Correction Notes

Public sanitization:

- Removed original cover-page team names, student numbers, and course-submission identity.
- Replaced private author metadata with a public pseudonym.
- No private email, phone number, address, local path, token, or credential was detected in the PDF text scan.

Technical corrections applied:

- Corrected the m2 equation sign convention to: $m_2 \ddot{x}_2 - k_1 x_1 + (k_1 + k_2) x_2 - k_2 x_3 = 0$.
- Corrected the total equation matrix vector so the stiffness matrix multiplies $[x_1, x_2, x_3]^T$.

Remaining limitations:

- Numerical screenshots were preserved from the original report; regenerate them from the cleaned source code for stricter reproducibility.
- This is an educational simulator/report and is not certified for safety-critical or production engineering use.

Technical corrections (October 2026, checked against the simulation model):

- Corrected peak displacements to match the model: k_0 table and text (0.32 / 0.26 / 0.20 m), k_3 table and text (0.28 / 0.20 m), heavy tip $m_3 = 2$ kg (1.9 cm), excitation table (0.26 m).
- $m_3 = 0.5$ kg: the tip has the smallest peak (2.2 cm vs 2.6 cm at x_1); the 'whipping' claim and the derived design advice were corrected. $m_3 = 2$ kg: only x_2 grows (2.1 -> 2.5 cm).
- Reference state: $x_1 \sim x_3$, not $x_1 \sim x_2$ (Tables 1 and 2); peaks are 2.1-2.6 cm, not a uniform 2.4 cm.
- $k_1 = 2000$ N/m lowers x_1 (Table 5). $k_2 = 500$ N/m lowers x_3 to 0.21 m while x_2 peaks at 0.27 m.
- $k_3 = 2000$ N/m leaves x_1 practically unchanged (0.254 m vs 0.255 m) instead of raising it.
- Excitation location: the equal chain is symmetric, so tip excitation mirrors root excitation (same 0.26 m peak); the actuator advice now names either end.
- Viscous damping (Table 15): about 20 cycles in 0-5 s for every c ; 6 % ($c = 0.5$) and 1 % ($c = 1$) of the energy remain at $t = 5$ s.
- Coulomb friction: the amplitude decays linearly, so the energy follows a parabola; the smoothed tanh friction model has no true stiction.
- Impulse duration: $E = I^2/2m$ follows from the delta-v conversion and does not test the RK4 solver.
- Determinant expansion: removed a duplicated $(-k_1)$ factor and an extra bracket.
- Labels: springs $k_0..k_3$ in the layout line, 'Term 3 (c_2)', 'TIP MASS SENSITIVITY (m_3)', coupling-spring subject lines, Table 5 states renamed S36/S37 (S9/S10 were already used), figures 45-48.
- Fabrication: q_1 Case 1 is $f = 1.5/0.6667 = 2.25$ Hz (14.14 rad/s); q_1 Case 2 interval is 0.785714 s; the Case 2 impulse is on M2; Case 2 q_1 is lower, not higher, than Case 1.
- With the measured k and m the model predicts 3.13 / 6.18 / 8.54 Hz (Case 1); the measured 1.5-2.25 Hz lie below the lowest mode, so the agreement with the model is qualitative only.